CALCULUS

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Basics (pre-calculus):

Logarithms: If $\log_b a = m$ then $a = b^m$

1. $\log_c(ab) = \log_c a + \log_c b$
2. $\log_c(a/b) = \log_c a - \log_c b$
3. $\log_{b^n} a^m = \frac{m}{n}\log_b a$
4. $\log_b a = \frac{1}{\log_a b} = \log_c a \cdot \log_b c = \frac{\log_c a}{\log_c b}$
5. $a^x = e^{\log_e a^x} = e^{x\log_e a}$ $\qquad \log_e \rightarrow \ln$
6. $\log_a a = 1$, $\log_a 1 = 0$, $\log_c(1/a) = -\log_c a$

Ex: $\log_2 5 = \frac{\log_{10} 5}{\log_{10} 2} = 2.322$

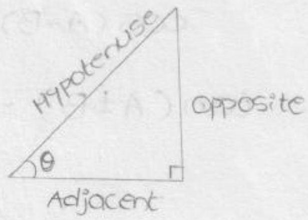
GATE IN(08): The expression $e^{-\ln x}$ for $x>0$ is equal to

Sol: $e^{-\ln x} = e^{-\log_e x}$

$$= e^{\log_e(1/x)} = 1/x = x^{-1}$$

Trignometry:

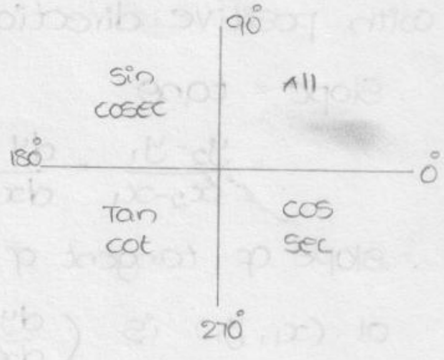
$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{\text{cosec}\,\theta}$$



$$\cos\theta = \frac{\text{Adjacent}}{\text{hypotenuse}} = \frac{1}{\sec\theta}$$

$$\tan\theta = \frac{\text{opposite}}{\text{Adjacent}} = \frac{\sin\theta}{\cos\theta} = \frac{1}{\cot\theta}$$

	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
$\cos\theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
$\tan\theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞



(i) If the angle is $(90\pm\theta)$, $(270\pm\theta)$ i.e

about vertical line, then co-function of initial function will apply.

(ii) If the angle is $(180\pm\theta)$, $(360\pm\theta)$ i.e. about horizontal line then same function will apply.

(iii) The positivity or negativity of the resulting function depends on the quadrant in which <u>initial</u> <u>function</u> lies.

Ex: $\sin(270+\theta) = -\cos\theta$ $\qquad$ $\cot(270-\theta) = \tan\theta$

$\cos(90+\theta) = -\sin\theta$ $\qquad$ $\sec(180+\theta) = -\sec\theta$

Formulas:

1. $\sin^2\theta + \cos^2\theta = 1$; $\operatorname{cosec}^2\theta - \cot^2\theta = 1$; $1+\tan^2\theta = \sec^2\theta$.

2. $\sin(A\pm B) = \sin A\cdot\cos B \pm \cos A\cdot \overset{\sin}{\cancel{\cos}}B$

   put $B=A \Rightarrow \sin 2A = 2\sin A\cos A$

   $\sin(A+B) + \sin(A-B) = 2\sin A\cos B$

   $\sin(A+B) - \sin(A-B) = 2\cos A\sin B$

3. $\cos(A\pm B) = \cos A\cos B \mp \sin A\sin B$

   put $B=A$ $\quad \cos 2A = \cos^2 A - \sin^2 A$

   $= 1 - 2\sin^2 A = 2\cos^2 A - 1$

   $\cos(A+B) + \cos(A-B) = 2\cos A\cos B$

   $\cos(A-B) - \cos(A+B) = 2\sin A\sin B$

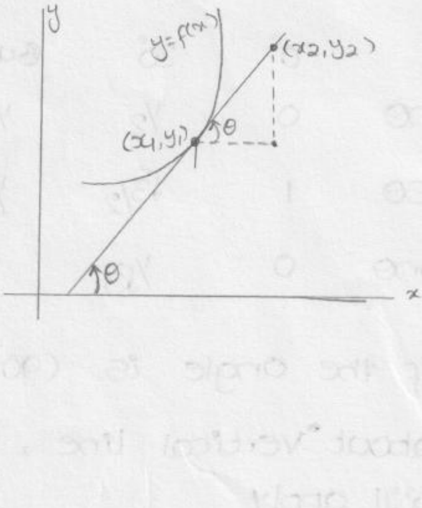
4. $\tan(A\pm B) = \dfrac{\tan A \pm \tan B}{1\mp \tan A\cdot\tan B}$

   put $B=A \Rightarrow \tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

Slope of a line: It is the tangent of angle made by the line with positive direction of x-axis.

Slope $= \tan\theta$

$= \dfrac{y_2-y_1}{x_2-x_1} = \dfrac{dy}{dx}$



slope of tangent of curve ~~at~~ $y=f(x)$ at (x_1,y_1) is $\left(\dfrac{dy}{dx}\right)_{(x_1,y_1)}$

Equation of straight line:

1. point slope form : point is (x_1, y_1) & slope m

$$y - y_1 = m(x - x_1)$$

2. Two point form: $y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

3. slope intercept form: slope is 'm', y-intercept is 'c'

$$y = mx + c$$

4. Intercept form: x intercept is 'a', 'y' intercept is 'b'

$$\frac{x}{a} + \frac{y}{b} = 1$$

5. General equation of straight line $ax + by + c = 0$

$$y = \left(\frac{-a}{b}\right)x + \left(\frac{-c}{b}\right)$$

slope $= -a/b$ y-intercept $-c/b$

$$ax + by = -c$$

$$\frac{x}{(-c/a)} + \frac{y}{(-c/b)} = 1 \Rightarrow x\text{-intercept} = -c/a$$

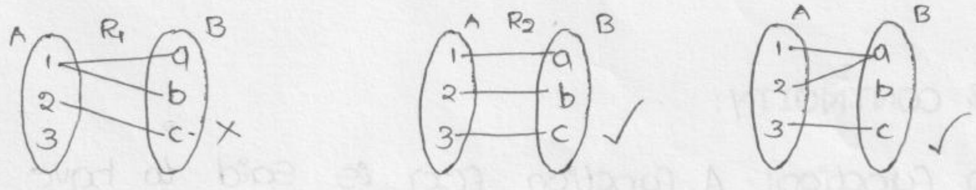
Function: A function from set A to set B is a relation from A to B satisfying the condition

"To each element in A, there exist a unique element in B"

i.e (i) every element in A must be associated

(ii) It must be associated with a unique element in B.

Ex:



Explicit function: $z = f(x_1, x_2, \ldots x_n)$

Dependent variable — Independent variable

Implicit function: $\phi(z, x_1, x_2, \ldots x_n) = c$

composit function: If $z = f(x, y)$ where $x = \phi(t)$ & $y = \psi(t)$ i.e

z is a function of some function.

Some Special functions:

1. Even function: $f(-x) = f(x)$ Ex: x^2, $\cos x$ ...

2. odd function: $f(-x) = -f(x)$ Ex: x, $\sin x$ --

3. Modulus function:

$$f(x) = |x| = \begin{cases} x & ; x>0 \\ -x & ; x<0 \\ 0 & ; x=0 \end{cases}$$

GATE-99: $f(x) = e^{x}$ neither even nor odd

4. Step / Greatest integer function:

$$f(x) = [x] = n \in Z \quad \text{where } n \le x \le n+1$$

Ex: $[7.2] = 7$; $[7.999] = 7$; $[-1.2] = -2$

Symmetric properties of the curve: Let $f(x,y) = c$ be the eq^n of a curve

(i) If $f(x,y)$ contains only even powers of x i.e $f(-x,y) = f(x,y)$ then it is symmetric about y-axis

(ii) If $f(x,y)$ contains only even powers of y i.e $f(x,-y) = f(x,y)$ then it is symmetric about x-axis

(iii) If $f(x,y) = f(y,x)$ then it is symmetric about $y=x$.

GATE (97): The curve given by the equation $x^2+y^2 = 3axy$ is

(a) symmetrical about x-axis (b) y-axis (c) line $y=x$

(d) tangential to $x=y=a/3$

Sol: $f(x,y) = f(y,x) \Rightarrow x^2+y^2 = 3axy$

$\therefore$ Symmetrical about line $y=x$.

LIMITS & CONTINUITY:

Limit of a function: A function $f(x)$ is said to have limit value 'l' as x tend to 'a' if

$$\lim_{x\to a} f(x) = l$$

Left limit: when $x<a$, $x\to a$ $\quad \lim_{x\to a^-} f(x) = \lim_{h\to 0} f(a-h)$

Right limit: when $x>a$, $x\to a$ $\quad \lim_{x\to a^+} f(x) = \lim_{h\to 0} f(a+h)$

A limit exists if LHL = RHL

Ex: If $f(x) = \begin{cases} 2x+3 & \text{for } x \geq 2 \\ x+9 & \text{for } x<2 \end{cases}$ then $\lim\limits_{x \to 2} f(x) = ?$

sol: $\lim\limits_{x \to 2^+} 2x+3 = 7 \neq \lim\limits_{x \to 2^-} x+9 = 11$

$\therefore$ Limit doesnot exist.

Indeterminate form: $\frac{0}{0}, \frac{\infty}{\infty}, 0\times\infty, \infty-\infty, 0^0, 1^\infty, \infty^0$

Standard limits:

(i) $\lim\limits_{x \to a} \frac{x^n - a^n}{x-a} = na^{n-1}$

(ii) $\lim\limits_{x \to 0} \frac{e^{mx}-1}{x} = e$

(iii) $\lim\limits_{x \to 0} \frac{a^x-1}{x} = \log_e a$

(iv) $\lim\limits_{x \to 0} [1+ax]^{1/x} = e^a$

(v) $\lim\limits_{x \to 0} [1+\frac{a}{x}]^x = e^a$

(vi) $\lim\limits_{x \to 0} \frac{\sin x}{x} = 1$ or $\lim\limits_{x \to 0} \frac{\tan x}{x} = 1$

(vii) $\lim\limits_{x \to \infty} \frac{\sin x}{x} = 0$

(viii) $\lim\limits_{x \to 0} \frac{\sin mx}{x} = m$

(ix) $\lim\limits_{x \to 0} \left[\frac{a^x+b^x}{2}\right]^{1/x} = \sqrt{ab}$

(x) $\lim\limits_{x \to 0} [\cos x + a\sin bx]^{1/x} = e^{ab}$

(xi) $\lim\limits_{x \to 0} \left[\frac{1-\cos ax}{x^2}\right] = \frac{a^2}{2}$

(xii) $\lim\limits_{x \to \infty} \frac{\log_e x}{x} = 0$

Limit properties:

1. $\lim\limits_{x \to a} [c\, f(x)] = c \lim\limits_{x \to a} f(x)$

2. $\lim\limits_{x \to a} [f(x) \pm g(x)] = \lim\limits_{x \to a} f(x) \pm \lim\limits_{x \to a} g(x)$

3. $\lim\limits_{x \to a} [f(x) \cdot g(x)] = \lim\limits_{x \to a} f(x) \cdot \lim\limits_{x \to b} g(x)$

4. $\lim\limits_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim\limits_{x \to a} f(x)}{\lim\limits_{x \to a} g(x)}$ provided $\lim\limits_{x \to a} g(x) \neq 0$

5. $\lim\limits_{x \to a} [f(x)]^n = \left[\lim\limits_{x \to a} f(x)\right]^n$ where 'n' is a real number.

6. $\lim\limits_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim\limits_{x \to a} f(x)}$

7. $\lim\limits_{x \to a} c = c$; c is a real number

8. If $P(x)$ is a polynomial then $\lim\limits_{x \to a} P(x) = P(a)$

(GATE-95): $\underset{x\to 0}{lt}\ x\sin(1/x) =$ ______.

sol: Let $x = t \Rightarrow 1/x = 1/t$, $t \to \infty$

$$\underset{x\to 0}{lt}\ x\sin(1/x) = \underset{t\to\infty}{lt}\ \frac{\sin t}{t} = 0$$

(GATE-99): $\underset{n\to\infty}{lt}\ \frac{n}{\sqrt{n^2+n}} =$

sol: $$\underset{n\to\infty}{lt}\ \frac{n}{\sqrt{n^2+n}} = \underset{n\to\infty}{lt}\ \frac{n}{n\sqrt{1+1/n}}$$

$$= \underset{n\to\infty}{lt}\ \frac{1}{\sqrt{1+1/n}} = \frac{1}{\sqrt{1+0}} = 1$$

(GATE-01 IN): $\underset{x\to\pi/4}{lt}\ \frac{\sin 2(x-\pi/4)}{x-\pi/4}$

sol: Let $x - \pi/4 = t \Rightarrow t \to 0$

$$\underset{x\to\pi/4}{lt}\ \frac{\sin 2(x-\pi/4)}{x-\pi/4} = \underset{t\to 0}{lt}\ \frac{\sin 2t}{t} = 2$$

(GATE-02 CE): $\underset{n\to\infty}{lt}\ n^{1/n} =$

sol: Let $y = \underset{n\to\infty}{lt}\ n^{1/n}$ Apply log on b.s

$$\log_e y = \underset{n\to\infty}{lt}\ \log_e n^{1/n} = \underset{n\to\infty}{lt}\ \frac{1}{n}\log_e n = 0$$

$$\log_e y = 0 \Rightarrow y = e^0 = 1$$

(GATE-03): $\underset{x\to 0}{lt}\ \frac{\sin^2 x}{x} = \underline{0}$

sol: $$\underset{x\to 0}{lt}\ \frac{\sin^2 x}{x} = \underset{x\to 0}{lt}\ \frac{\sin x}{x} \times \sin x$$

$$= \underset{x\to 0}{lt}\ \frac{\sin x}{x} \times \underset{x\to 0}{lt}\ \sin x$$

$$= 1 \times 0 = 0$$

(GATE-04): $\underset{x\to 0}{lt}\ \frac{x^3+x^2}{2x^3-7x^2} =$

sol: $$\underset{x\to 0}{lt}\ \frac{x^3+x^2}{2x^3-7x^2} = \underset{x\to 0}{lt}\ \frac{x^2(x+1)}{x^2(2x-7)}$$

$$= \underset{x\to 0}{lt}\ \frac{x+1}{2x-7} = \frac{1}{-7} = -\frac{1}{7}$$

(GATE-07 ME): $\underset{x\to 0}{lt}\ \frac{e^x - (1+x+x^2/2)}{x^3}$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

by substituting we will get

$$\lim_{x\to 0} \frac{\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots+\infty}{x^3} = \lim_{x\to 0}\left(\frac{1}{3!}+\frac{x}{4!}+\frac{x^2}{5!}+\cdots+\infty\right)$$

$$= \frac{1}{3!}+0 = \frac{1}{6}$$

(GATE-07 EC): $\lim_{\theta\to 0} \frac{\sin(\theta/2)}{\theta}$ is

Sol: $\lim_{x\to 0} \frac{\sin mx}{x} = m \Rightarrow \lim_{\theta\to 0} \frac{\sin(\frac{1}{2}\cdot\theta)}{\theta} = \frac{1}{2} = 0.5$

(GATE-08 CS): $\lim_{x\to\infty} \frac{x-\sin x}{x+\cos x} =$

Sol: $$= \lim_{x\to\infty} \frac{1-\frac{\sin x}{x}}{1+\frac{\cos x}{x}} = \frac{1-\lim_{x\to\infty}\frac{\sin x}{x}}{1+\lim_{x\to\infty}\frac{\cos x}{x}} = \frac{1-0}{1+0} = 1$$

(GATE-08 ME): $\lim_{x\to 8} \frac{x^{1/3}-2}{x-8}$

Sol: $$= \lim_{x\to 8} \frac{x^{1/3}-8^{1/3}}{x-8} \qquad \lim_{x\to a}\frac{x^n-a^n}{x-a} = n\cdot a^{n-1}$$

$$= \frac{1}{3}\cdot 8^{\frac{1}{3}-1} = \frac{1}{3}\times 8^{-2/3} = \frac{1}{3}\times\frac{1}{4} = \frac{1}{12}$$

(GATE-08 PI): $\lim_{x\to 0} \frac{\sin x}{e^x\cdot x} =$

Sol: $$\lim_{x\to 0} \frac{\sin x}{e^x\cdot x} = \lim_{x\to 0}\frac{\sin x}{x}\times\lim_{x\to 0}\frac{1}{e^x}$$

$$= 1\times\frac{1}{e^0} = 1\times 1 = 1$$

(GATE-10 CS): $\lim_{n\to\infty}(1-1/n)^{2n} =$

Sol: $$\lim_{n\to\infty}(1-1/n)^{2n} = \lim_{n\to\infty}\left[\left(1+\frac{-1}{n}\right)^n\right]^2$$

$$= \left[\lim_{n\to\infty}\left(1+\frac{(-1)}{n}\right)^n\right]^2 = (e^{-1})^2 = e^{-2}$$

(GATE-14 CE): $\lim_{x\to\infty} \frac{x+\sin x}{x} =$

Sol: $$\lim_{x\to\infty}\frac{x}{x} + \underbrace{\lim_{x\to\infty}\frac{\sin x}{x}}_{0} = \lim_{x\to\infty} 1+0 = 1+0 = 1$$

Pblm: $\lim_{x\to 0}\frac{|x|}{x} = ?$

Sol: $$\text{LHL} = \lim_{x\to 0^-}\frac{|x|}{x} = \lim_{h\to 0}\frac{|-h|}{-h} = -1$$

$$\text{RHL} = \lim_{x\to 0^+}\frac{|x|}{x} = \lim_{h\to 0}\frac{|h|}{h} = 1$$

LHL $\neq$ RHL $\therefore$ limit doesn't exist.

Continuity of $f(x)$: A function $f(x)$ is said to be continous at $x=a$, when $\underset{x \to a}{\text{lt}} f(x) = f(a)$

Ex: $f(x) = \begin{cases} 2x+3 & \text{for } x>2 \\ x+5 & \text{for } x<2 \\ 15 & \text{for } x=2 \end{cases}$

$$\underset{x \to 2^+}{\text{lt}} f(x) = 7 = \underset{x \to 2^-}{\text{lt}} f(x) = 7 \neq f(2) = 15$$

Limit exists but not continous.

(GATE-97): If $y=|x|$ for $x<0$ and $y=x$ for $x \geq 0$ then

√(a) $\frac{dy}{dx}$ is discontinous at $x=0$ (b) y is discontinous at $x=0$

(c) y is not defined at $x=0$ (d) Both y & dy/dx are discontinous at $x=0$

Sol: $y=|x|$ is continous everywhere

$\frac{dy}{dx} = \frac{|x|}{x}$ is not continous at $x=0$ because $\underset{x \to 0}{\text{lt}} \frac{|x|}{x}$ = not exist

$\therefore$ Ans: (a)

(GATE-2013 CS): which of the following function is continous at $x=3$

√(a) $f(x) = \begin{cases} 2 & \text{if } x=3 \\ x-1 & \text{if } x>3 \\ \frac{x+3}{3} & \text{if } x<3 \end{cases}$ (b) $f(x) = \begin{cases} 4 & \text{if } x=3 \\ 8-x & \text{if } x \neq 3 \end{cases}$

(c) $f(x) = \begin{cases} x+3 & \text{if } x \leq 3 \\ x-4 & \text{if } x>3 \end{cases}$ (d) $f(x) = \frac{1}{x^3-27}$ if $x \neq 3$

Sol:
(a) $\underset{x \to 3^+}{\text{lt}} f(x) = 3-1 = 2$ $\underset{x \to 3^-}{\text{lt}} f(x) = \frac{3+3}{3} = 2$ $f(3) = 2$

$\therefore f(x)$ is continous at $x=3$

Ans: (a)

Note: If a function is continous at $x=a$ then limit of the function also exists at $x=a$ and is equal to $f(a)$.

Differentiability: A function $f(x)$ is said to be differentiable at $x=a$ if $\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ exists and is denoted $f'(a)$

In general the derivative function is given by

$$f'(x) = \lim_{\Delta x\to 0} \frac{f(x+\Delta x)-f(x)}{\Delta x}$$

Ex: If $f(x) = x^2$ then $f'(x) = 2x$ & $f'(x) = 2a$

$$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a} = \lim_{x\to a} \frac{x^2-a^2}{x-a} = \lim_{x\to a} x+a = 2a$$

$$f'(x) = \lim_{\Delta x\to 0} \frac{f(x+\Delta x)-f(x)}{\Delta x} = \frac{(x+\Delta x)^2-x^2}{\Delta x}$$

$$= \lim_{\Delta x\to 0} \frac{x^2+\Delta x^2+2\cdot x\cdot\Delta x-x^2}{\Delta x} = \lim_{\Delta x\to 0} \frac{\Delta x(\Delta x+2x)}{\Delta x}$$

$$= \lim_{\Delta x\to 0} \Delta x+2x = 0+2x = 2x$$

* Left hand derivative: $LHD = \lim_{h\to 0} \frac{f(c-h)-f(c)}{-h}$

Right hand derivative: $RHD = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h}$

Necessary condition for function to be differential is

$$LHD = RHD$$

* Every differentiable function is a continous function. But every continous function is not differentiable

Note:

(i) $|x|$ is not differentiable at $x=0$

$$LHD = \lim_{h\to 0} \frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0} \frac{|-h|-0}{-h} = \lim_{h\to 0} \frac{h}{-h} = -1$$

$$RHD = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} \frac{|h|-0}{h} = \lim_{h\to 0} \frac{h}{h} = 1$$

$LHD \neq RHD \Rightarrow |x|$ not differentiable.

(ii) $|x-a|$ is not differentiable at $x=a$

(iii) $|ax+b|$ is not differentiable at $x=-b/a$

(GATE-95): The function $f(x) = |x+1|$ on the interval $[-2,0]$

Sol: $|x+a|$ is continous everywhere.

$|x+a|$ is not differentiable at $x=-a$.

$\therefore |x+1|$ is not differentiable at $x=-1 \in [-2,0]$

Ans: continous, but not differentiable.

(GATE-07 IN): The function $f(x) = |x|^3$, at $x=0$ is = (where x is real)

(a) continous but not differentiable

(b) once differentiable but not twice

✓(c) twice " " thrice (d) thrice differentiable

Sol:

$$f(x) = |x|^3 = \begin{cases} x^3 ; & x>0 \\ -x^3 ; & x<0 \\ 0 ; & x=0 \end{cases}$$

$f(0)=0$, LHL = RHL = 0 $\Rightarrow$ continous.

$$LHD = \lim_{h\to0} \frac{f(0-h)-f(0)}{-h} = \lim_{h\to0} \frac{|-h|^3-0}{-h} = \lim_{h\to0} \frac{-h^3}{-h} = \lim_{h\to0} h^2 = 0$$

$$RHD = \lim_{h\to0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to0} \frac{|h|^3-0}{-h} = \lim_{h\to0} -h^2 = 0$$

LHD = RHD $\Rightarrow$ differentiable.

$$f'(x) = \begin{cases} 3x^2 ; & x>0 \\ -3x^2 ; & x<0 \\ 0 ; & x=0 \end{cases} \qquad f''(x) = \begin{cases} 6x ; & x>0 \\ -6x ; & x<0 \\ 0 ; & x=0 \end{cases} \qquad f'''(x) = \begin{cases} 6 ; & x>0 \\ -6 ; & x<0 \\ \text{Not diff'ble} ; & x=0 \end{cases}$$

$\therefore f(x)$ is twice differentiable but not thrice.

(GATE-10 ME): The function $y = |2-3x|$

Sol:

$y = |2-3x|$ continous $\forall x \in R$

$y = |2-3x|$ not differentiable at $x = 2/3$

$2-3x=0$

$x = 2/3$

Derivatives of Some functions:

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^n	$\to nx^{n-1}$	e^x	$\to e^x$	$\sec x$	$\to \sec x \cdot \tan x$
$\sqrt{x}$	$\to \frac{1}{2\sqrt{x}}$	$\log_e^x$	$\to 1/x$	$\operatorname{cosec} x$	$\to -\operatorname{cosec} x \cdot \cot x$
$\frac{1}{x^n}$	$\to \frac{-n}{x^{n+1}}$	$\sin x$	$\to \cos x$	$\sin^{-1}x$ (or) $-\cos^{-1}x$	$\to \frac{1}{\sqrt{1-x^2}}$
$\frac{1}{\sqrt{x}}$	$\to \frac{-1}{2x\sqrt{x}}$	$\cos x$	$\to -\sin x$	$\tan^{-1}x$ (or) $-\cot^{-1}x$	$\to \frac{1}{1+x^2}$
a^x	$\to a^x \log_e a$	$\tan x$	$\to \sec^2 x$	$\sec^{-1}x$ (or) $-\operatorname{cosec}^{-1}x$	$\to \frac{1}{x\sqrt{x^2-1}}$
		$\cot x$	$\to -\operatorname{cosec}^2 x$		

Note: (1) $(uv)' = uv' + u'v$

(2) $\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{v^2}$

(3) $\frac{d}{dx}[f(g(x))] = f'(g(x)).g'(x)$

(4) Leibniz formula: $(f.g)^n = nc_0 f^n g + nc_1 f^{n-1} g' + \dots + nc_n f.g^n$

where 'n' is order of differentiation.

L'Hospital rule: If $\lim_{x\to a} \frac{f(x)}{g(x)}$ takes the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)} = \frac{f'(a)}{g'(a)}$$

Note: 1. In case of other indeterminate forms $0.\infty, 0^0, \infty^0, 1^\infty, \infty-\infty$, first we have to reduce it to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ and use this rule

2. For the forms $0^0, \infty^0, 1^\infty$ take logarithm and then take limits

3. $0^\infty = 0$, $\infty^\infty = \infty$, $\infty.\infty = \infty$, $\infty+\infty = \infty$, $\infty^{-\infty} = 0$ are not indeterminate

Ex: $\lim_{x\to 0} \frac{\ln(1+x)}{\sin x} = \frac{0}{0}$ form

Apply L'H rule $= \lim_{x\to 0} \frac{1/(1+x)}{\cos x} = \frac{1}{1} = 1$

Ex: $\lim_{x\to 0} x^x = 0^0$ form Let $y = x^x$

Apply logarithm first $\ln y = \ln x^x = x \ln x$

$$\lim_{x\to 0} \ln y = \lim_{x\to 0} x.\ln x$$

$$\ln\left[\lim_{x\to 0} y\right] = \lim_{x\to 0} \frac{\ln x}{1/x} = \frac{\infty}{\infty} \text{ Apply L'H}$$

$$= \lim_{x\to 0} \frac{1/x}{-1/x^2} = -\lim_{x\to 0} x = 0$$

$$\lim_{x\to 0} y = \lim_{x\to 0} x^x = e^0 = 1$$

(GATE-93 ME): $\lim_{x\to 0} \frac{x(e^x-1) + 2(\cos x - 1)}{x(1-\cos x)}$ $\frac{0}{0}$

Sol: using L'H rule

$$\lim_{x\to 0} \frac{(e^x-1) + x.e^x + 2(-\sin x)}{(1-\cos x) + x(\sin x)} \quad \frac{0}{0}$$

Again apply L'H rule.

$$= \lim_{x\to 0} \frac{e^x + e^x + xe^x - 2\cos x}{\sin x + \sin x + x\cos x} \quad \frac{0}{0}$$

Again apply L'H rule

$$= \lim_{x\to 0} \frac{e^x + e^x + e^x + xe^x + 2\sin x}{\cos x + \cos x + \cos x - x\sin x} = \frac{1+1+1+0}{1+1+1-0} = 1$$

(GATE-95 CS): $\lim_{x\to\infty} \frac{x^3 - \cos x}{x^2 + (\sin x)^2} \quad \frac{\infty}{\infty}$

Sol: Apply L' Hospital

$$\lim_{x\to\infty} \frac{3x^2 + \sin x}{2x + 2\sin x \cdot \cos x} = \lim_{x\to\infty} \frac{3x + \frac{\sin x}{x}}{2 + \frac{\sin 2x}{x}} = \frac{3\times\infty + 0}{2+0} = \infty$$

(GATE-99 IN): $\lim_{x\to 0} \frac{1}{10} \cdot \frac{1 - e^{-j5x}}{1 - e^{-jx}} \quad \frac{0}{0}$

Sol: Apply L' Hospital

$$\lim_{x\to 0} \frac{1}{10} \cdot \frac{-(-5j)e^{-j5x}}{-(-j) \cdot e^{-jx}} = \frac{5j}{10j} = 1/2$$

(GATE-99): $\lim_{x\to a} (x-a)^{x-a} \quad 0^0$

Sol: Let $y = (x-a)^{(x-a)} \Rightarrow \log y = (x-a)\log(x-a)$

$$\lim_{x\to a} \log y = \lim_{x\to a} [(x-a)\log(x-a)]$$

$$\log\left[\lim_{x\to a} y\right] = \lim_{x\to a} \frac{\log(x-a)}{1/(x-a)} \quad \frac{\infty}{\infty}$$

Apply L'H rule $= \lim_{x\to a} \frac{1/x-a}{-1/(x-a)^2} = \lim_{x\to a} -(x-a) = 0$

$$\lim_{x\to a} y = \lim_{x\to a} (x-a)^{(x-a)} = e^0 = 1$$

(GATE-07 PI): $\lim_{x\to \pi/4} \frac{\cos x - \sin x}{x - \pi/4} \quad \frac{0}{0}$

Sol: Apply L'H rule $= \lim_{x\to\pi/4} \frac{-\sin x - \sin x}{1} = \frac{-1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\sqrt{2}$

(GATE-12 ME, PI): $\lim_{x\to 0} \left(\frac{1-\cos x}{x^2}\right) \quad \frac{0}{0}$

Sol: Apply L'H rule $= \lim_{x\to 0} \frac{\sin x}{2x} \quad \frac{0}{0}$

Again apply L'H $= \lim_{x\to 0} \frac{\cos x}{2} = \frac{1}{2}$

(GATE-14 ME): $\lim_{x\to 0} \frac{x - \sin x}{1 - \cos x} \quad \frac{0}{0}$

Sol: Apply L'H rule

$= \underset{x \to 0}{lt} \dfrac{1-\cos x}{\sin x} \quad \dfrac{0}{0}$.

again apply L'H rule $= \underset{x \to 0}{lt} \dfrac{\sin x}{\cos x} = 0$

(GATE-14 ME): $\underset{x \to 0}{lt} \left(\dfrac{e^{2x}-1}{\sin(4x)} \right) \quad \dfrac{0}{0}$

Sol: Apply L'H $= \underset{x \to 0}{lt} \dfrac{2 \cdot e^{2x}}{4 \cdot \cos 4x} = \dfrac{2}{4} = \dfrac{1}{2}$

** (GATE-14 CE): $\underset{a \to 0}{lt} \dfrac{x^a - 1}{a} = \dfrac{0}{0}$

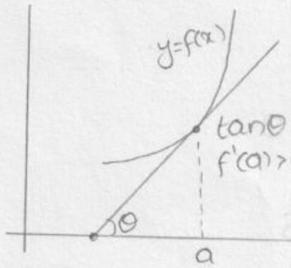
Sol: Apply L'H differentiate w.r.to a

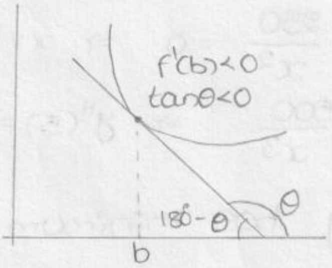
$= \underset{a \to 0}{lt} \dfrac{x^a \cdot \log x - 0}{1} = x^0 \cdot \log x = \log x$.

Increasing (or) decreasing nature of function:

A function $y = f(x)$ will have increasing nature at $x = a$, if $f'(a) > 0$

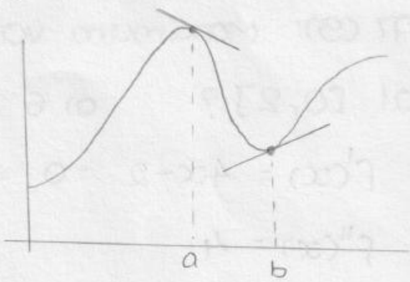
A function $y = f(x)$ will have decreasing nature at $x = b$ if $f'(b) < 0$





Local
Maxima and minima:

To obtain the maxima or minima



1. Find $f'(x)$ and solve the equation $f'(x) = 0$ to obtain the stationary points say $x = a, b, c$.
2. Find $f''(x)$
3. If $f''(a) < 0$, then at $x = a$. $f(x)$ will have maximum value

   If $f''(b) > 0$, then at $x = b$. $f(x)$ will have minimum value

   If $f''(c) = 0$, then at $x = c$. $f(x)$ won't have minimum/maximum. The point is called saddle point.

Absolute maxima/minima: If $f(x)$ defined in interval $[a, b]$ then

Absolute minimum value $= \min \{ f(a), f(b), \text{all local minimum values} \}$

Absolute maximum value $= \max \{ f(a), f(b), \text{all local maximum values} \}$

If stationary points are out of given interval then don't consider them

Ex: The function $f(x) = x^3 - 9x^2 + 24x - 12$ has

Sol: $f'(x) = 3x^2 - 18x + 24 = 0$ $\qquad f''(x) = 6x - 18$

$x = 2, 4$ $\qquad f''(2) = -6 < 0$

$f''(4) = 6 > 0$

$\therefore f(x)$ has Max. at $x = 2$, Min at $x = 4$.

(GATE-07 EC)

Ex: The maximum value of $f(x) = x^2 - x - 2$ in the interval $[-4, 4]$ is

✓(a) 18 (b) 10 (c) -2.25 (d) indeterminate

Sol: $f'(x) = 2x - 1 = 0 \Rightarrow x = 1/2$

$f''(x) = 2$ can't find

$f(-4) = 18$ $\quad f(4) = 10$ $\quad f(1/2) = -2.25$

$\therefore f(x)$ has maximum at $x = -4$.

(GATE-94): The function $y = x^2 + \frac{250}{x}$ at $x = 5$ attains

Sol: $y' = 2x - \frac{250}{x^2} = 0 \Rightarrow x = 5$

$y'' = 2 + \frac{500}{x^3} \Rightarrow y''(5) = 2 + 4 = 6 > 0$

$\therefore$ y has minimum at $x = 6$

(GATE-97 CS): Maximum value of function $f(x) = 2x^2 - 2x + 6$ in the interval $[0, 2]$? a) 6 ✓b) 10 c) 12 d) 5.5

$\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad f(1/2)$

Sol: $f'(x) = 4x - 2 = 0 \Rightarrow x = 1/2$

$f''(x) = 4$

$f(0) = 0 + 6 = 6$ $\qquad f(2) = 8 - 4 + 6 = 10$

$\therefore$ Maximum value $= 10$ in $[0, 2]$

(GATE-99 CE): Number of inflection points for the curve $y = 2x^4 + x$

Sol: $\frac{d^2y}{dx} = 0 \Rightarrow 8 \times 3x^2 = 0$

$\Rightarrow x = 0 \Rightarrow y = 0$

$\therefore$ (0,0) is the only inflection point.

(GATE-05 EE): For the function $f(x) = x^2 e^{-x}$, the max. occurs when $x = \underline{?}$

Sol: $f'(x) = e^{-x}[-x^2 + 2x] = 0 \Rightarrow x = 0, 2$ $\qquad f''(0) = 2 > 0$

$f''(x) = -e^{-x}[-x^2 + 2x] + e^{-x}[-2x + 2]$ $\qquad f''(2) = -2e^{-2} < 0$

$\therefore f(x)$ has max. at $x = 2$

(GATE-07 EE): The function $f(x) = (x^2-4)^2$ has (x is real number)

(a) only one minimum ✓(b) only two minima (c) Three minima

(d) Three maxima.

Sol: $f'(x) = 2(x^2-4) \cdot 2x = 0 \Rightarrow x = \pm 2, 0$

$f''(x) = 4(3x^2-4)$

$f''(0) = -16 < 0$ $\quad f''(2) = 32 > 0$ $\quad f''(-2) = 32 > 0$

$\therefore f(x)$ has minimum at $x = \pm 2$

(GATE-08 CS): The number of distinct extrema for the curve $3x^4 - 16x^3 + 24x^2 + 37$ is

Sol: $f'(x) = 12x^3 - 48x^2 + 48x = 0 \Rightarrow x = 0, 2, 2$

$f''(x) = 36x^2 - 96x + 48$

$f''(0) = 48 > 0$ $\quad f''(2) = 0$ No extrema

$\therefore f(x)$ has <u>only one</u> extrema (minimum) at $x = 0$

(GATE-08 EC): For real values of 'x', the minimum value of function $f(x) = e^x + e^{-x}$ is

Sol: $f(x) = e^x + e^{-x}$ where $x \in R$

$f'(x) = e^x - e^{-x} = 0 \Rightarrow e^x = e^{-x} \Rightarrow x = 0$

$f''(x) = e^x + e^{-x} \Rightarrow f''(0) = 1 + 1 = 2 > 0$

$\therefore f(x)$ has Minimum at $x = 0$, Minimum value $f(0) = 2$

(GATE-10 EC): If $e^y = x^{1/x}$ then y has a

Sol: $e^y = x^{1/x} \Rightarrow y = \log x^{1/x} = \dfrac{\log x}{x}$

$y' = \dfrac{1 - \log x}{x^2} = 0 \Rightarrow x = e$

$y'' = \dfrac{x^2(-1/x) - (1 - \log x)x}{x^4}$

$y''(e) = -e/e^4 < 0 \Rightarrow y$ has maximum at $x = e$

(GATE-12 EC, EE, IN): The maximum value of $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$

Similar Ques. in GATE-14 EC

Sol: $f'(x) = 3x^2 - 18x + 24 = 0 \Rightarrow x = 2, 4$

$f''(x) = 6x - 18$ $\quad f''(2) = -6 < 0$ $\quad f'(4) = 6 > 0$

↳ local maximum at $x = 2$

Local maximum = $f(2) = 25$.

In the interval $[1,6]$ $\quad f(1) = 21 \quad f(6) = 41$

$\therefore$ Absolute maximum = $\max\{f(1), f(6), \text{local max.}\}$

$= \max\{21, 41, 25\} = 41.$

(GATE-14 EC): For $0 \le t < \infty$ the maximum value of the function $f(t) = e^{-t} - 2e^{-2t}$ occurs at

Sol: $f'(t) = (-e^{-t} + 4e^{-2t}) = 0 \Rightarrow e^{-t} = 1/4$

$\Rightarrow t = \log_e 4$

$f''(t) = (e^{-t} - 8e^{-2t})$

$f''(\log_e 4) = e^{-\log_e 4} - 8e^{-2\log_e 4} = \frac{1}{4} - 8 \cdot \frac{1}{16} = -1/4 < 0$

$\therefore f(t)$ has maximum value at $t = \log_e 4$

(GATE-14 EC): The max value of function $f(x) = \ln(1+x) - x$ (where $x > -1$) occurs at $x =$

Sol: $f'(x) = \frac{1}{1+x} - 1 = 0 \Rightarrow x = 0$

$f''(x) = \frac{-1}{(1+x)^2} \Rightarrow f''(0) = -1 < 0$

$\therefore f(x)$ has maximum at $x = 0$

(GATE-14 EE): The max. value of the function $f(x) = xe^{-x}$ in the interval $(0, \infty)$ is

Sol: $f'(x) = (-x \cdot e^{-x} + e^{-x}) = 0 \Rightarrow x = 1$

At $x = 1$, $f''(x) < 0$

$\therefore$ Maximum exists at $x = 1$ and $f(1) = 1 \cdot e^{-1} = 1/e$

(GATE-14 EE): Minimum of the real valued function $f(x) = (x-1)^{2/3}$ occurs at $x =$

Sol: $f(x) = \left[(x-1)^{1/3}\right]^2 \quad \therefore f(x) \ge 0$ always

The minimum value of $f(x)$, $f(x) = 0$

$(x-1)^{2/3} = 0 \Rightarrow x = 1$

(GATE-14 EE): The minimum value of $f(x) = x^3 - 3x^2 - 24x + 100$ in interval $[-3, 3]$ is

Sol: $f'(x) = 3x^2 - 6x - 24 = 0 \Rightarrow x = -2, 4$

but $x = 4 \notin [-3, 3]$

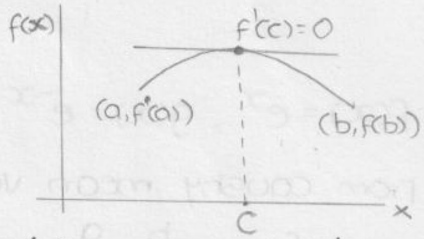
$f''(x) = 6x - 6 \Rightarrow f''(-2) = -18 < 0$ we get maximum

$\therefore$ Absolute $\overset{in}{\cancel{max}}$imum $= \min\{f(-3), f(3)\}$ $\quad f(-3) = 118 \quad f(3) = 28$

$= \min\{118, 28\} = 28$

MEAN VALUE THEOREMS:

Rolle's theorem: If $f(x)$ is a function defined in an interval $[a,b]$ such that

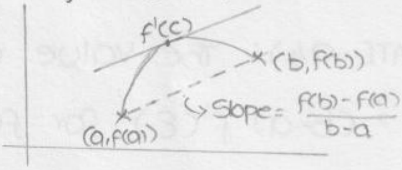


(1) $f(x)$ is continous in $[a,b]$

(2) $f'(x)$ exists in (a,b)

(3) $f(a) = f(b)$ then there exist $c \in (a,b)$ such that $f'(c) = 0$.

Langranges theorem: If $f(x)$ is a function defined in an interval $[a,b]$ such that



(1) $f(x)$ is continous in $[a,b]$

(2) $f'(x)$ is exists in (a,b) then there exist $c \in (a,b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b-a} \qquad \min_{a \le x \le b} f(x) \le \frac{f(b)-f(a)}{b-a} \le \max_{a \le x \le b} f'(x)$$

cauchy's mean value theorem: If $f(x)$, $g(x)$ are two functions defined in an interval such that

(1) $f(x)$, $g(x)$ are continous in $[a,b]$

(2) $f'(x)$, $g'(x)$ exists in (a,b)

(3) $g'(x) \neq 0 \ \forall \ x \in (a,b)$ then there exist $c \in (a,b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

prob: The value 'c' of mean value theorem for $f(x) = x^2 - 5x + 6$ in the interval $[2,3]$ is

sol: $f(2) = 4 - 10 + 6 = 0$

$f(3) = 9 - 15 + 6 = 0$

From rolle's theorem $\exists \ c \in (2,3)$ such that $f'(c) = 0$

$\Rightarrow 2c - 5 = 0 \Rightarrow c = 5/2$

prob: $f(x) = x^3 - 4x^2 + 4x$ in the interval $[0,2]$ is

$f(0) = 0 \qquad f(2) = 8 - 16 + 8 = 0$

$\therefore$ From rolle's theorem $\exists \ c \in (0,2)$ such that $f'(c) = 0$

$3c^2-8c+4=0 \Rightarrow c=2,\ 2/3 \qquad c=2 \notin (0,2)$

$c=2/3 \in (0,2)$

prob: $f(x)=1+x^2$ in $[1,2]$ is

$f(1)=2 \qquad f(2)=5 \qquad f(a)\neq f(b)$

from langranges theorem $f'(c)=\frac{5-2}{1}=3$

$2c=3 \Rightarrow c=3/2 \in (1,2)$

prob: $f(x)=e^x,\ g(x)=e^{-x}$ in $[a,b]$

sol: From caushy mean value theorem

$$\frac{e^c}{-e^{-c}}=\frac{e^b-e^a}{e^{-b}-e^{-a}} \Rightarrow -e^{2c}=-e^{a+b}$$

$$\Rightarrow 2c=a+b \Rightarrow c=\frac{a+b}{2} \in (a,b)$$

(GATE-94): The value of ε in the mean value theorem of $f(b)-f(a) = (b-a)f'(\varepsilon)$ for $f(x)=Ax^2+Bx+C$ in (a,b) is

sol: $f'(\varepsilon)=\frac{f(b)-f(a)}{b-a}$

$$2A\varepsilon+B=\frac{Ab^2+Bb+C-Aa^2-Ba-C}{b-a}$$

By comparing

$$2A\varepsilon+B=A(b+a)+B \qquad 2\varepsilon=a+b \Rightarrow \varepsilon=\frac{a+b}{2}$$

(GATE-95): If $f(0)=2$ and $f'(x)=\frac{1}{5-x^2}$, then the lower and upper bounds of $f(1)$ estimated by the mean value theorem are

sol: Let $f(x)$ be defined in $[0,1]$ by lagrange's M.V.T

$\exists\, c\in(0,1)$ such that $f'(c)=\frac{f(1)-f(0)}{1-0}$

$$\frac{1}{5-c^2}=\frac{f(1)-2}{1}$$

We know that $\text{Min}\{f'(x)\} < f'(c) < \text{Max}\{f'(x)\} \qquad c\in(0,1)$

$$\frac{1}{5} < f(1)-2 < \frac{1}{4} \Rightarrow 2.2 < f(1) < 2.25$$

TAYLOR SERIES: The Taylor series of $f(x)$ about $x=a$ is given by

$$f(x)=f(a)+\frac{(x-a)}{1!}f'(a)+\frac{(x-a)^2}{2!}f''(a)+\frac{(x-a)^3}{3!}f'''(a)+\cdots$$

Maclaurin's Series: put $a=0$

$$f(x)=f(0)+x\cdot f'(0)+\frac{x^2}{2!}f''(0)+\frac{x^3}{3!}f'''(0)+\cdots$$

Important expansions:

1. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

2. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$

3. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

4. $\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$

5. $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \cdots$

6. $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \cdots$

prob: In the taylor series of $f(x) = e^x$ about $x=2$ the co.eff of $(x-2)^4$ is

Sol:
$$= \frac{1}{4!} f^{IV}(2) = \frac{1}{4!} e^x \Big|_{x=2} = \frac{e^2}{24}$$

prob: In the taylor series of $f(x) = e^x + \sin x$ about $x-\pi$ the co.eff of $(x-\pi)^2$ is

Sol:
$$= \frac{1}{2!} f''(\pi) = \frac{1}{2!}\left[e^x - \sin x\right]\Big|_{x=\pi} = \frac{e^\pi}{2}$$

prob: The linear approximation for e^{-x} around $x=2$ is

Sol:
$$f(x) = f(2) + (x-2) f'(2)$$
$$= e^{-2} + (x-2)(-e^{-2}) = e^{-2}(3-x)$$

(GATE-00 CE): The Taylor series expansion of $\sin x$ about $x = \pi/6$ is

Sol:
$$f(x) = f(a) + (x-a) f'(a) + \frac{(x-a)^2}{2!} f''(a) + \cdots$$

$f(x) = \sin x$ and $a = \pi/6 \Rightarrow \sin(\pi/6) = 1/2 = f(a)$

$f'(x) = \cos x \Rightarrow f'(a) = \cos \pi/6 = \sqrt{3}/2$

$f''(x) = -\sin x \Rightarrow f''(a) = -\sin \pi/6 = -1/2$

$$\sin x = \frac{1}{2} + (x - \pi/6)\sqrt{3}/2 + \frac{(x-\pi/6)^2}{2!}\left(-\frac{1}{2}\right) + \cdots$$
$$= \frac{1}{2} + \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right) - \frac{1}{4}\left(x - \frac{\pi}{6}\right)^2 + \cdots$$

(GATE-01 CE): Limit of the following series as x approaches $\pi/2$ is $f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sin x$

Sol:
$$\lim_{x\to\pi/2} \sin x = \sin \pi/2 = 1$$

(GATE-09 EC): The Taylor series expansion of $\frac{\sin x}{x-\pi}$ at $x=\pi$ is given by

Sol: $$\frac{\sin x}{x-\pi} = \frac{1}{x-\pi}\left[\sin\pi + (x-\pi)(\cos\pi) + \frac{(x-\pi)^2}{2!}(-\sin\pi) + \cdots\right]$$

$$= \frac{1}{x-\pi}\left[0 + (x-\pi)(-1) + \frac{(x-\pi)^2}{2!}(0) + \frac{(x-\pi)^3}{3!} - \cdots\right]$$

$$= -1 + \frac{(x-\pi)^2}{3!} - \frac{(x-\pi)^4}{5!}$$

(or)

put $x-\pi = t \Rightarrow x = \pi + t$

$$f = \frac{\sin(\pi+t)}{t} = -\frac{\sin t}{t} \quad \text{about } t=0$$

$$= -\frac{1}{t}\left[t - \frac{t^3}{3!} + \frac{t^5}{5!} \cdots\right] = -1 + \frac{t^2}{3!} - \frac{t^4}{5!} + \cdots$$

$$= -1 + \frac{(x-\pi)^2}{3!} - \frac{(x-\pi)^4}{5!} \cdots$$

(GATE-14 EC): The Taylor series expansion of $3\sin x + 2\cos x$ is

Sol: $$3\sin x + 2\cos x = 3\left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right] + 2\left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right]$$

$$= 2 + 3x - x^2 - \frac{x^3}{2} + \cdots$$

PARTIAL DIFFERENTIATION:

If $f(x,y)$ is a function of two variables (x,y) then

$$\frac{\partial f}{\partial x} = \lim_{\Delta x\to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$

Ex: $f(x) = x^3 + 3xy^2$

$\frac{\partial f}{\partial x} = 3x^2 + 3y^2$ $\quad$ $\frac{\partial f}{\partial y} = 3x(2y) = 6xy$

$\frac{\partial^2 f}{\partial x^2} = 6x$ $\quad$ $\frac{\partial f^2}{\partial y^2} = 6x$

$\frac{\partial^2 f}{\partial y.\partial x} = 6y$ $\quad$ $\frac{\partial^2 f}{\partial x.\partial y} = 6y$

Euler's theorem: If $f(x,y)$ is a homogenous function of degree 'n' then

$$x\cdot\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = nf$$

$$x^2\frac{\partial^2 f}{\partial x^2} + 2xy\frac{\partial^2 f}{\partial x.\partial y} + y^2\frac{\partial^2 f}{\partial y^2} = n(n-1)f$$

Homogeneous function: A function $f(x,y)$ is said to be a homogeneous function of degree 'n' if

$f(x,y) = x^n \cdot \phi(y/x)$ (or) $y^n \varphi(x/y)$.

Ex: 1. $f(x,y) = x^3 + 3xy^2 = x^3\left[1+3\left(y/x\right)^2\right]$ deg = 3

2. $f(x,y) = \log_e y - \log_e x = x^0 \log_e\left(y/x\right)$ deg = 0

3. $f(x,y) = y \sin(y/x) = y \sin\left(\frac{1}{x/y}\right)$ deg = 1

4. $f(x,y) = \sin\left(\frac{x^4+y^4}{x-y}\right)$ Not homogeneous

Note: If $u(x,y) = f(x,y) + g(x,y) + h(x,y)$, where f, g & h are homogenous function with degree m, n & p respectively then

(a) $x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = mf + ng + ph$

(b) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \cdot \partial y} + y^2 \cdot \frac{\partial^2 u}{\partial y^2} = m(m-1)f + n(n-1)g + p(p-1)h$

Note: If f(u) is homogenous function in two variables x & y with degree 'n', then

(a) $x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = n \frac{f(u)}{f'(u)} = F(u)$.

(b) $x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \cdot \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = F(u)[F'(u) - 1]$

Total differentiation: If $z = f(x,y)$ where $x = \phi(t)$, $y = \psi(t)$, then the total derivative of 'z' w.r.to 't' is

$$\frac{\partial z}{\partial t} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

Total differential co.efficient $dz = \frac{\partial f}{\partial x} \cdot dx + \frac{\partial f}{\partial y} \cdot dy$

Note: 1. If $f(x,y) = C$ is an implicit function then $\frac{dy}{dx} = \frac{-f_x}{f_y}$

(2) If $z = f(x,y)$ where $x = \phi(u,v)$ & $y = \psi(u,v)$ then

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial v}$$

pblm: If $z = e^x \sin y$, where $x = \log t$ & $y = t^2$ then $\frac{dz}{dt} = ?$

Sol:
$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$x = \log t$
$t = e^x$

$$= e^x \sin y \cdot \frac{1}{t} + e^x \cos y \cdot 2t$$

$$= \frac{e^x}{t}(\sin y + 2t^2 \cos y) = \sin y + 2y \cos y$$

prob: The total derivative of x^2y w.r.to 'x', where x & y are connected by the relation $x^2+xy+y^2=1$ is

sol: Let $u=x^2y \Rightarrow \frac{du}{dx} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dx} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dx}$

$$= 2xy + \cancel{y} x^2 \, dy/dx$$

Given relation $x^2+xy+y^2=1 = f(x,y)$ say

$$\frac{dy}{dx} = \frac{-f_x}{f_y} = -\frac{2x+y}{x+2y}$$

$$\therefore \frac{du}{dx} = 2xy + x^2\left(-\frac{2x+y}{x+2y}\right) = 2xy - x^2\left(\frac{2x+y}{x+2y}\right)$$

prob: If $u = f(x+cy) + g(x-cy)$ then $\frac{u_{xx}}{u_{yy}} =$

sol: Let $r = x+cy \qquad s = x-cy$

$$\frac{\partial r}{\partial x} = 1 \quad \frac{\partial r}{\partial y} = c \qquad \frac{\partial s}{\partial x} = 1 \quad \frac{\partial s}{\partial y} = -c$$

$$u = f(r) + g(s) \Rightarrow u_x = f'(r)\cdot\frac{\partial r}{\partial x} + g'(s)\cdot\frac{\partial s}{\partial x} = f'(r) + g'(s)$$

$$u_{xx} = f''(r) + g''(s)$$

$$u_y = f'(r)\cdot\frac{\partial r}{\partial y} + g'(s)\cdot\frac{\partial s}{\partial y} = f'(r)\cdot c + g'(s)(-c)$$

$$\therefore u_{yy} = f''(r)c^2 + g''(s)\cdot c^2$$

$$\frac{u_{xx}}{u_{yy}} = \frac{1}{c^2} = c^{-2}$$

prob: If $\mu = \frac{x^2y}{x^{5/2}+y^{5/2}}$ then $x^2\mu_{xx} + 2xy\,\mu_{xy} + y^2\mu_{yy} =$

sol:

$$\mu = \frac{x^2y}{x^{5/2}+y^{5/2}} = \frac{x^2y}{x^2\left[x^{1/2}+y^{1/2}\right]} = \frac{y}{x^{1/2}+y^{1/2}}$$

$$= \frac{y}{y^{1/2}\left[\left(\frac{x}{y}\right)^{1/2}+1\right]} = y^{1/2}\cdot\frac{1}{\left[1+\left(x/y\right)^{1/2}\right]}$$

$$\therefore n = 1/2$$

$$x^2\mu_{xx} + 2xy\,\mu_{xy} + y^2\mu_{yy} = n(n-1)\mu = \frac{1}{2}\left(\frac{1}{2}-1\right)\mu$$

$$= -\frac{1}{4}\mu.$$

prob: If $u = \operatorname{cosec}^{-1}\left[\frac{x^{1/4}-y^{1/4}}{x^{1/5}+y^{1/5}}\right]$ then $x u_x + y u_y =$

sol: $\operatorname{cosec} u = \frac{x^{1/4}-y^{1/4}}{x^{1/5}+y^{1/5}} \qquad n = \frac{1}{4}-\frac{1}{5} = \frac{1}{20}$

$$x u_x + y u_y = n\frac{f(u)}{f'(u)} = \frac{1}{20}\times\frac{\operatorname{cosec} u}{-\operatorname{cosec} u\cdot\cot u} = -\frac{1}{20}\tan u.$$

(GATE-08 ME): Let $f = y^x$ what is $\frac{\partial^2 f}{\partial x \cdot \partial y}$ at $x=2,\ y=1$?

Sol: $f = y^x \Rightarrow \frac{\partial f}{\partial y} = x \cdot y^{x-1}$ (x const)

$$\frac{\partial}{\partial x} \cdot \frac{\partial f}{\partial y} = y^{x-1} + x \cdot y^{x-1} \cdot \log y \cdot (1) \quad (y \text{ const})$$

$$\left.\frac{\partial^2 f}{\partial x \partial y}\right|_{\substack{x=2\\y=1}} = y^{x-1} + x \cdot y^{x-1} \log y = 1^1 + 2 \cdot 1^1 \log 1 = 1$$

Maxima & Minima for function of two variables:

Let $z = f(x,y)$

consider $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$, $r = \frac{\partial^2 z}{\partial x^2}$, $s = \frac{\partial^2 z}{\partial x \cdot \partial y}$, $t = \frac{\partial^2 z}{\partial y^2}$

Method: (i) Find p, q, r, s, t.

(ii) Equate p & q to zero for obtaining the stationary points

(iii) At each stationary point find r, s, t.

(a) If $rt - s^2 > 0$, $r > 0$ then the function $f(x,y)$ has a minima at the stationary point

(b) If $rt - s^2 > 0$, $r < 0$ then $f(x,y)$ has a maximum at the stationary point

(c) If $rt - s^2 < 0$, then $f(x,y)$ has no extreme at that point and is known as saddle point.

Constrained Maxima & minima:

Lagranges method of undetermined multipliers:

Let $f(x,y,z)$ & $\phi(x,y,z) = C$ then $F(x,y,z) = f(x,y,z) + \lambda \phi(x,y,z) = 0$

$$\left.\begin{array}{lll} F_x = 0 & \text{i.e} & \frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0 \\ F_y = 0 & \text{i.e} & \frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0 \\ F_z = 0 & \text{i.e} & \frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \end{array}\right\} \text{Lagrange's eq}^{n}\text{s}$$

λ - lagranges multiplier.

(GATE-93 ME): The function $f(x,y) = x^2 y - 3xy + 2y + x$ has

Sol: $\frac{\partial f}{\partial x} = 2xy - 3y + 1 = 0$ $\qquad \frac{\partial f}{\partial y} = x^2 - 3x + 2 = 0 \Rightarrow x = 1, 2$

at $x=1 \Rightarrow 2(1)y-3y+1=0 \Rightarrow y=1 \Rightarrow (1,1)$

at $x=2 \Rightarrow 2(2)y-3y+1=0 \Rightarrow y=-1 \Rightarrow (2,-1)$

$$\frac{\partial^2 f}{\partial x^2}=2y \Rightarrow \left.\frac{\partial^2 f}{\partial x^2}\right|_{(1,1)}=2 \qquad \frac{\partial f^2}{\partial x\cdot\partial y}=2x-3$$

$$\frac{\partial^2 f}{\partial y^2}=0$$

$$\left[\frac{\partial^2 f}{\partial x^2}\times\frac{\partial^2 f}{\partial y^2}\right]-\left[\frac{\partial^2 f}{\partial x\cdot\partial y}\right]^2<0 \text{ at both } (1,1) \,\&\, (2,-1)$$

$\therefore$ NO extremum

(GATE-02): The function $f(x,y)=2x^2+2xy-y^3$ has stationary points

sol: $\frac{\partial f}{\partial x}=4x+2y=0 \qquad \frac{\partial f}{\partial y}=2x-3y^2=0$

$\Rightarrow 2x=-y \qquad \Rightarrow -y-3y^2=0 \Rightarrow y=0, -1/3$

$\therefore x=0, 1/6$

$\therefore (0,0), (1/6, -1/3)$ are the stationary points

(GATE-07 PI): For the function $f(x,y)=x^2-y^2$ defined on R^2, the point $(0,0)$ is

sol: $\frac{\partial f}{\partial x}=2x=0 \Rightarrow x=0 \qquad \frac{\partial f}{\partial y}=-2y=0 \Rightarrow y=0$

$(x,y)=(0,0)$

$$\frac{\partial^2 f}{\partial x^2}=2 \qquad \frac{\partial^2 f}{\partial y^2}=-2 \qquad \frac{\partial^2 f}{\partial x\,\partial y}=0$$

$$\frac{\partial^2 f}{\partial x^2}\cdot\frac{\partial^2 f}{\partial y^2}-\left(\frac{\partial^2 f}{\partial x\cdot\partial y}\right)^2=2(-2)=-4<0$$

$\therefore$ Neither maxima nor minima

prob: If $u=\log_e\left(\frac{x^4+y^4}{x+y}\right)$ then $x\cdot\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=$

sol: $e^u=\frac{x^4+y^4}{x+y}=P$ say $\qquad n=3$

$$x\frac{\partial P}{\partial x}+y\frac{\partial P}{\partial y}=nP \Rightarrow x\cdot e^u\cdot\frac{\partial u}{\partial x}+y\cdot e^u\cdot\frac{\partial u}{\partial y}=ne^u$$

$$\Rightarrow x\cdot\frac{\partial u}{\partial x}+y\cdot\frac{\partial u}{\partial y}=3$$

Prob: The minimum value of $x^2+y^2+z^2$ where $x+y+z=1$ is

sol: $f=x^2+y^2+z^2 \qquad \phi=x+y+z-1 \qquad x+y+z=1$

$F=f+\lambda\phi \qquad x=1/3 \quad y=1/3 \quad z=1/3$

$$\left.\begin{array}{l} f_x=2x+\lambda=0 \Rightarrow \lambda=-2x \\ f_y=2y+\lambda=0 \Rightarrow \lambda=-2y \\ f_z=2z+\lambda=0 \Rightarrow \lambda=-2z \end{array}\right\} \Rightarrow x=y=z$$

$f_{min}=x^2+x^2+x^2$
$=3x^2=3(1/3)=1$

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Integration: If $\frac{d}{dx} f(x) = F(x)$ then $\int F(x) = F(x) + C$

$f(x)$	$\int f(x)$	$f(x)$	$\int f(x)$
x^n	$\frac{x^{n+1}}{n+1}$	$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1}x$ (or) $-\cos^{-1}x$
$1/x$	$\log_e x$	$\frac{1}{1+x^2}$	$\tan^{-1}x$ (or) $-\cot^{-1}x$
$1/x^2$	$-1/x$	$\frac{1}{x\sqrt{x^2-1}}$	$\sec^{-1}x$ (or) $-\operatorname{cosec}^{-1}x$
$\sqrt{x}$	$\frac{2}{3}x\sqrt{x}$	$\frac{1}{\sqrt{a^2-x^2}}$	$\sin^{-1}(x/a)$
$1/\sqrt{x}$	$2\sqrt{x}$	$\frac{1}{\sqrt{x^2-a^2}}$	$\cosh^{-1}(x/a)$
e^x	e^x	$\frac{1}{\sqrt{a^2+x^2}}$	$\sinh^{-1}(x/a)$
a^x	$a^x/\log_e a$		
$\sin x$	$-\cos x$		
$\cos x$	$\sin x$		
$\sec^2 x$	$\tan x$		
$\operatorname{cosec}^2 x$	$-\cot x$		
$\sec x \cdot \tan x$	$\sec x$		
$\operatorname{cosec} x \cdot \cot x$	$-\operatorname{cosec} x$		

$* \int \frac{f'(x)}{f(x)} dx = \log_e |f(x)|$ *

$\int \tan x \, dx = \int \frac{\sin x}{\cos x} dx = -\log_e \cos x = \log_e |\sec x|$

$\int \cot x \, dx = \int \frac{\cos x}{\sin x} dx = \log_e |\sin x|$

$\int \sec x \, dx = \int \frac{\sec x \cdot (\sec x + \tan x)}{\sec x + \tan x} dx = \log_e |\sec x + \tan x|$

$\int \operatorname{cosec} x \, dx = \log_e |\operatorname{cosec} x - \cot x|$

Integration by parts:

ILATE

$\int uv \, dx = u \int v dx - \int u' \left(\int v dx\right) dx$

$= uv_1 - u'v_2 + u''v_3 - u'''v_4 + \cdots$

Ex: $\int \log_e x \, dx = \log_e x \int dx - \int \frac{1}{x} \cdot x \, dx = x(\log_e x - 1)$

$\int x^2 e^x dx = x^2 e^x - 2x e^x + 2e^x = e^x(x^2 - 2x + 2)$

$\int x e^x dx = x e^x - e^x = e^x(x-1)$

$$\int \frac{1}{1-\cos x}dx = \int \frac{(1+\cos x)}{(1-\cos x)(1+\cos x)}dx$$

$$= \int \frac{(1+\cos x)}{1-\cos^2 x}dx = \int \frac{1+\cos x}{\sin^2 x}dx$$

$$= \int \operatorname{cosec}^2 x\,dx + \int \frac{\cos x}{\sin x}\cdot\frac{1}{\sin x}dx$$

$$= -\cot x + \int \cot x\cdot \operatorname{cosec} x\,dx = -\cot x - \operatorname{cosec} x$$

prob: $\int \frac{x^4}{x^2+1}dx = \int \frac{x^4-1+1}{x^2+1}dx$

$$= \int \frac{(x^2+1)(x^2-1)+1}{x^2+1}dx = \int (x^2-1)dx + \int \frac{1}{x^2+1}dx$$

$$= \frac{x^3}{3} - x + \tan^{-1}x$$

prob: $\int \frac{\sin^8 x}{\cos^{10} x}dx = \int \tan^8 x\cdot \sec^2 x\,dx$

Let $\tan x = t \Rightarrow \sec^2 x\,dx = dt$

$$= \int t^8\,dt = \frac{t^9}{9} = \frac{\tan^9 x}{9}$$

prob: $\int \frac{\sin^2 x}{1+\cos x}dx = \int \frac{1-\cos^2 x}{1+\cos x}dx$

$$= \int \frac{(1-\cos x)\cdot(1+\cos x)}{1+\cos x}dx = \int (1-\cos x)dx$$

$$= x - \sin x$$

prob: $\int \frac{x^2-a^2}{x^2+a^2}dx = \int \frac{x^2+a^2-2a^2}{x^2+a^2}dx$

$$= \int dx - 2a^2\int \frac{1}{x^2+a^2}dx = x - 2a^2\cdot\frac{1}{a}\cdot\tan^{-1}(x/a)$$

$$= x - 2a\tan^{-1}(x/a)$$

prob: $\int \frac{x}{\sqrt{1-x^2}}dx =$

Let $1-x^2 = t \Rightarrow -2x\,dx = dt \Rightarrow x\,dx = \frac{-dt}{2}$

$$\int \frac{x}{\sqrt{1-x^2}}dx = \int \frac{1}{\sqrt{t}}\cdot\left(\frac{-dt}{2}\right) = \frac{-1}{2}\int \frac{1}{\sqrt{t}}dt$$

$$= -\frac{1}{2}\cdot 2\sqrt{t} = -\sqrt{t} = -\sqrt{1-x^2}$$

prob: $\int \frac{1}{x^2+2x+2}dx = \int \frac{1}{(x+1)^2+1}dx$

Let $x+1 = t \Rightarrow dx = dt$

$$= \int \frac{1}{t^2+1}dt = \tan^{-1}t = \tan^{-1}(1+x)$$

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DEFINITE INTEGRALS: If $\int f(x)\,dx = F(x)$ then $\int_a^b f(x)\,dx = F(b) - F(a)$

Note: $\frac{d}{dx}\left[\int_{u(x)}^{v(x)} f(x)\,dx\right] = f(v)\cdot\frac{dv}{dx} - f(u)\cdot\frac{du}{dx}$

properties of definite integral:

(i) $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

(ii) If $c \in (a,b)$ then $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

(iii) $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

(iv) $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}\,dx = \frac{b-a}{2}$

(v) $\int_{-a}^a f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & ; \text{ If } f(x) \text{ is even} \\ 0 & ; \text{ If } f(x) \text{ is odd} \end{cases}$

(vi) $\int_0^{2a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & ; \text{ If } f(2a-x) = f(x) \\ 0 & ; \text{ If } f(2a-x) = -f(x) \end{cases}$

(vii) $\int_0^a x\cdot f(x)\,dx = \frac{a}{2}\int_0^a f(x)\,dx$; If $f(a-x) = f(x)$

(viii) $\int_0^{na} f(x)\,dx = n\int_0^a f(x)\,dx$; If $f(x+a) = f(x)$

(ix) $\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = \left[\frac{n-1}{n}\times\frac{n-3}{n-2}\times\cdots\times\frac{2}{3} \text{ (or) } \frac{1}{2}\right]K$

where $K = \begin{cases} 1 & ; \text{ if } n \text{ is } \cancel{\text{even}} \text{ odd} \\ \pi/2 & ; \text{ if } n \text{ is } \cancel{\text{odd}} \text{ even} \end{cases}$

(x) $\int_0^{\pi/2} \sin^m x\cdot\cos^n x\,dx = \left\{\frac{[(m-1)(m-3)\cdots 2 \text{ (or) } 1][(n-1)(n-3)\cdots 2 \text{ (or) } 1]}{[(m+n)(m+n-2)(m+n-4)\cdots]}\right\}\cdot K$

where $K = \begin{cases} \pi/2 & ; \text{ when } m \,\&\, n \text{ are even} \\ 1 & ; \text{ Otherwise} \end{cases}$

Prob: $\int_{-2\pi}^{2\pi} \cos^6 x\,dx$

Sol: $f(x) = \cos^6 x$ is a even function $\int_{-2\pi}^{2\pi} \cos^6 x\,dx = 2\int_0^{2\pi} \cos^6 x\,dx$

$f(2\pi - x) = [\cos(2\pi - x)]^6 = \cos^6 x$

$\int_{-2\pi}^{2\pi} \cos^6 x\,dx = 2\int_0^{2\pi} \cos^6 x\,dx = 2\cdot 2\int_0^{\pi} \cos^6 x\,dx$

$$= 4 \cdot 2\int_0^{\pi/2} \cos^6 x \, dx \qquad \cos^6(\pi - x) = [-\cos x]^6 = \cos^6 x$$

$$= 8 \times \frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{5\pi}{4}$$

prob: $\int_1^3 x\sqrt{x^2-1}\, dx$

Let $x^2 - 1 = t^2$ $\quad x=1 \Rightarrow t=0$

$2x\,dx = 2t\,dt$ $\quad x = 3 \Rightarrow 3^2 - 1 = t^2$

$x\,dx = t\,dt$ $\quad t = \sqrt{8}$

$$= \int_0^{\sqrt{8}} t \cdot t\,dt = \int_0^{\sqrt{8}} t^2 dt$$

$$= \frac{t^3}{3}\Big|_0^{\sqrt{8}} = \frac{(\sqrt{8})^3}{3} = \frac{8\sqrt{8}}{3}$$

prob: $\int_0^1 x\cdot\frac{\sin^{-1}x}{\sqrt{1-x^2}}\, dx$

Let $\sin^{-1}x = t \Rightarrow x = \sin t$ $\quad x=0 \Rightarrow t = \sin^{-1}0 = 0$

$dx = \cos t\, dt$ $\quad x = 1 \Rightarrow t = \sin^{-1}1 = \pi/2$

$$= \int_0^{\pi/2} \sin t \cdot \frac{t}{\cos t}\cdot \cos t\, dt$$

$$= \int_0^{\pi/2} t\sin t\, dt = \left[t(-\cos t) - 1(-\sin t)\right]_0^{\pi/2}$$

$$= \left[-t\cos t + \sin t\right]_0^{\pi/2} = [1-0] = 1$$

prob: $\int_0^1 \sqrt{\frac{1-x}{1+x}}\, dx = \int_0^1 \sqrt{\frac{(1-x)(1-x)}{(1+x)(1-x)}}\, dx$

$$= \int_0^1 \frac{(1-x)}{\sqrt{1-x^2}}\, dx = \int_0^1 \frac{1}{\sqrt{1-x^2}}\, dx - \frac{1}{2}\int_0^1 \frac{2x}{\sqrt{1-x^2}}\, dx$$

$$= \left[\sin^{-1}x + \sqrt{1-x^2}\right]_0^1 = \sin^{-1}1 - \sqrt{1-0} = \frac{\pi}{2} - 1$$

prob: $\int_0^2 |1-x|\, dx = \int_0^1 (1-x)\, dx - \int_1^2 (1-x)\, dx$

$(1-x) > 0$ for $0 < x < 1$

$(1-x) < 0$ for $1 < x < 2$

$$= \left[x - \frac{x^2}{2}\right]_0^1 - \left[x - \frac{x^2}{2}\right]_1^2$$

$$= \left[1 - \frac{1}{2}\right] - \left[2 - \frac{4}{2} - 1 + \frac{1}{2}\right] = 1$$

prob: $\int_{-\pi}^{\pi} (|x| + \sin x)\, dx = \int_{-\pi}^{\pi} |x|\, dx + \int_{-\pi}^{\pi} \sin x\, dx$

$|x|$ even

$\sin x$ odd

$$= 2\int_0^{\pi} |x|\, dx + 0$$

$$= 2\left(x^2/2\right)_0^{\pi} = \pi^2$$

prob: $\int_0^{\pi/2} \frac{\tan x}{\tan x + \cot x}\, dx$

Sol: Let $f(x) = \tan x \Rightarrow f(0 + \pi/2 - x) = \tan\left(\frac{\pi}{2} - x\right) = \cot x$

$$\int_0^{\pi/2} \frac{\tan x}{\tan x + \cot x}\, dx = \int_0^{\pi/2} \frac{f(x)}{f(x) + f(0 + \pi/2 - x)} = \frac{\pi/2 - 0}{2}$$

$$= \pi/4$$

prob: $\int_0^n [x]\,dx =$

sol: $I = \int_0^1 0\,dx + \int_1^2 1\,dx + \int_2^3 2\,dx + \cdots$

$= (x]_1^2 + 2(x)_2^3 + 3(x)_3^4 + \cdots$

$= (2-1) + 2(3-1) + 3(4-3) + 4(5-4) + \cdots$

$= 1+2+3+4\cdots = \frac{n(n+1)}{2}$

prob: $\int_0^{\pi/2} \log(\tan x)\,dx$ is

sol: $\int_0^{\pi/2} \log(\tan x)\,dx = I$ say $\qquad \int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

$I = \int_0^{\pi/2} \log[\tan(\pi/2 - x)]\,dx = \int_0^{\pi/2} \log(\cot x)\,dx$

$I + I = \int_0^{\pi/2} [\log(\tan x) + \log(\cot x)]\,dx$

$2I = \int_0^{\pi/2} \log(\tan x \cdot \cot x)\,dx \qquad \tan x \cdot \cot x = 1$

$2I = \int_0^{\pi/2} \log(1)\,dx = 0 \Rightarrow I = 0$

prob: $\int_{-\pi}^{\pi} \log \frac{(1+\sin x)}{(1-\sin x)}\,dx = I$

$f(-x) = \log\left(\frac{1-\sin x}{1+\sin x}\right) = -\log\left(\frac{1+\sin x}{1-\sin x}\right) = -f(x)$

odd function $\therefore I = 0$

(GATE-97): If $\phi(x) = \int_0^{x^2} \sqrt{t}\,dt$ then $\frac{d\phi}{dx} =$

sol: $\phi = \frac{2}{3} t\sqrt{t}\Big|_0^{x^2} = \frac{2}{3} x^2\sqrt{x^2} - 0 = \frac{2}{3}x^3$

$\frac{d\phi}{dx} = \frac{2}{3} \cdot 3 \cdot x^2 = 2x^2$

(GATE-01): The value of the integral $I = \int_0^{\pi/4} \cos^2 x\,dx$

sol: $I = \int_0^{\pi/4} \cos^2 x\,dx = \int_0^{\pi/4} \left[\frac{1+\cos 2x}{2}\right] dx$

$= \frac{x}{2}\Big|_0^{\pi/4} + \frac{\sin 2x}{4}\Big|_0^{\pi/4} = \frac{\pi}{8} + \frac{1}{4} \cdot \sin(\pi/2)$

$= \frac{\pi}{8} + \frac{1}{4}$

(GATE-02): $\int_{-\pi/2}^{\pi/2} \frac{\sin 2x}{1+\cos x}\,dx = I$

$f(-x) = -f(x) \Rightarrow$ odd function $\therefore I = 0$

(GATE-05 IN): $\int_{-1}^{1} \frac{1}{x^2} dx$ $\qquad = 2\int_0^1 \frac{1}{x^2}dx = 2\cdot\left(-\frac{1}{x}\right)_0^1 = \infty$

Sol: $= \int_{-1}^{0} \frac{1}{x^2}dx + \int_{0}^{1}\frac{1}{x^2}dx = \left(\frac{x^{-1}}{-1}\right)_{-1}^{0} + \left(\frac{x^{-1}}{-1}\right)_{0}^{1}$

$= (\infty - 1) + (1 - \infty) = \infty$

(GATE-08 PI): $\int_{-\pi/2}^{\pi/2} (x\cos x)dx$

Sol: $f(x) = x\cos x \Rightarrow f(-x) = -x\cos(-x) = -x\cdot\cos x = -f(x)$

$\Rightarrow$ odd function $\quad \therefore I = 0$

(GATE-10 EE): $P = \int_0^1 x e^x dx$

Sol: $\int_0^1 x e^x dx = \left[xe^x - e^x\right]_0^1 = (e-e) + (e^0 - 0) = 1$

(GATE-11 CS): $\int_0^{\pi/2} \frac{\cos x + i\sin x}{\cos x - i\sin x} dx = I$

Sol: $I = \int_0^{\pi/2} \frac{e^{ix}}{e^{-ix}} dx = \int_0^{\pi/2} e^{2ix} dx = \left[\frac{e^{2ix}}{2i}\right]_{\pi/2}^{0}$

$= \frac{e^{i\pi}}{2i} - \frac{1}{2i} = \frac{\cos\pi + i(0)}{2i} - \frac{1}{2i} = -\frac{1}{2i} - \frac{1}{2i} = \frac{-2}{2i}$

$= -\frac{1}{i} = i$

(GATE-13 ME): $\int_1^e \sqrt{x}\cdot\ln(x)\cdot dx$ $\qquad \int u v = uv - \int v du$, $u = \ln x$

Sol: $= \left[\left(\ln x\cdot\frac{x^{3/2}}{3/2}\right)_1^e - \int_1^e \frac{1}{x}\cdot\frac{x^{3/2}}{3/2}dx\right]$

$= \frac{2}{3}\cdot e^{3/2} - 0 - \int_1^e \frac{2}{3}\cdot x^{1/2}dx$

$= \frac{2}{3}e^{3/2} - \frac{2}{3}\left(\frac{x^{3/2}}{3/2}\right)_1^e = \frac{2}{3}e^{3/2} - \frac{4}{9}e^{3/2} + \frac{4}{9} = \frac{2}{9}e^{3/2} + \frac{4}{9}$

(GATE-13 CE): $\int_0^{\pi/6} \cos^4 3\theta\cdot\sin^3 6\theta\cdot d\theta$

Sol: Let $3\theta = t \Rightarrow \theta = t/3 \Rightarrow d\theta = dt/3$ $\qquad \theta = 0 \Rightarrow t = 0$, $\theta = \pi/6 \Rightarrow t = \pi/2$

$I = \int_0^{\pi/2} \cos^4 t\cdot\sin^3 2t\cdot dt/3$

$= \frac{1}{3}\int_0^{\pi/2} \cos^4 t\cdot(2\sin t\cdot\cos t)^3 dt$

$= \frac{8}{3}\int_0^{\pi/2} \cos^7 t\cdot\sin^3 t\,dt = \frac{8}{3}\left[\frac{(6\times4\times2)(2)}{10\times8\times6\times4\times2}\right] = 1/15$

(GATE-14 ME): $\int_0^2 \frac{(x-1)^2\sin(x-1)}{(x-1)^2 + \cos(x-1)} dx$

Sol: $f(2-x) = \frac{(2-x-1)^2\sin(2-x-1)}{(2-x-1)^2 + \cos(2-x-1)} = \frac{(x-1)^2(-\sin(x-1))}{(x-1)^2 + \cos(x-1)} = -f(x)$

According to $\int_0^{2a} f(x)\,dx = 0$ if $f(2a-x) = -f(x)$

$\therefore I = 0$

Improper Integrals:

First Kind: $\int_a^b f(x)\,dx$ if $a=-\infty$ (or) $b=\infty$ (or) both

i.e $\int_{-\infty}^{b} f(x)\,dx$ or $\int_a^{\infty} f(x)\,dx$ (or) $\int_{-\infty}^{\infty} f(x)\,dx$

Second Kind: $\int_a^b f(x)\,dx$ if a & b are finite but $f(x)$ is infinite for some $x \in [a,b]$

Ex: $\int_{-1}^{1} \log(1+x)\,dx$, $\int_{-1}^{1} \sqrt{\frac{1+x}{1-x}}\,dx$; $\int_0^3 \frac{1}{x^2-5x+4}\,dx$

convergence of an improper integral:

(i) If $\int_a^b f(x)\,dx$ = finite, then it is a convergent improper integral

(ii) If $\int_a^b f(x)\,dx$ = Infinite, then it is a divergent improper integral

prob: $\int_0^{\infty} \frac{1}{a^2+x^2}\,dx$

sol: $I = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)\Big|_0^{\infty} = \frac{1}{a}\left[\tan^{-1}\infty - \tan^{-1}0\right]$

$= \frac{1}{a}\left[\frac{\pi}{2} - 0\right] = \frac{\pi}{2a}$ finite (convergent)

prob: $\int_{-\infty}^{0} e^{ax}\cos px\,dx$

sol: $\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}\left[a\cos bx + b\sin bx\right]$

$I = \left[\frac{e^{ax}}{a^2+p^2}(a\cos px + p\sin px)\right]_{-\infty}^{0} = \frac{a}{a^2+p^2} - 0 = \frac{a}{a^2+p^2}$ finite (convergent)

prob: $\int_{-1}^{1} \sqrt{\frac{1+x}{1-x}}\,dx$

sol: $I = \int_{-1}^{1} \sqrt{\frac{1+x}{1-x}} \times \frac{\sqrt{1+x}}{\sqrt{1+x}}\,dx = \int_{-1}^{1} \frac{1+x}{\sqrt{1-x^2}}\,dx$

$= \underbrace{\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\,dx}_{\text{even}} + \underbrace{\int_{-1}^{1} \frac{x}{\sqrt{1-x^2}}\,dx}_{\text{odd}}$

$= 2\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx + 0$

$= 2\sin^{-1}x\Big|_0^1 = 2\sin^{-1}1 = 2\cdot\pi/2 = \pi$ finite comparision test: For 1^{st} kind of improper integrals

(a) Let $0 \le f(x) \le g(x)$ then

(i) $\int_a^b f(x)dx$ converges if $\int_a^b g(x)$ is $\overset{conv}{\cancel{div}}$ergent.

(ii) $\int_a^b g(x)dx$ diverges if $\int_a^b f\cancel{g}(x)$ is divergent.

(b) Limit form: Let $f(x)$ & $g(x)$ be two positive functions such that $\lim_{x \to \infty} \frac{f(x)}{g(x)} = l$ (non-zero finite) then $\int_a^b f(x)dx$ & $\int_a^b g(x)dx$ both are converge or diverge together.

prob: $\int_1^\infty e^{-x^2} dx$

sol: $e^{x^2} \ge e^x \;\forall\; x \ge 1 \Rightarrow e^{-x^2} \le e^{-x} \;\forall\; x \le 1$

$\int_1^\infty e^{-x} dx = -e^{-x}\Big|_1^\infty = 1 \;\therefore$ convergent

$\therefore \int_1^\infty e^{-x^2} dx$ is also convergent.

prob: $\int_2^\infty \frac{1}{\log x} dx$

sol: $\log x < x \;\forall\; x \ge 2 \Rightarrow \frac{1}{\log x} > \frac{1}{x} \;\forall\; x \ge 2$

$\int_2^\infty \frac{1}{x} dx = \log x \Big|_2^\infty$ = infinite $\Rightarrow$ divergent

$\therefore \int_2^\infty \frac{1}{\log x} dx$ is also divergent.

Comparison test: For 2^{nd} kind of improper integrals

Limit form: Let $f(x)$ & $g(x)$ be two positive functions suchthat

(i) 'a' is a point of discontinuity and $\lim_{x \to a^+} \frac{f(x)}{g(x)} = l_1$ (non zero finite)

(ii) 'b' is a point of discontinuity and $\lim_{x \to b^-} \frac{f(x)}{g(x)} = l_2$ (non zero finite)

then $\int_a^b f(x)dx$ & $\int_a^b g(x)dx$ both converge (or) diverge together.

prob: $\int_1^2 \frac{\sqrt{x}}{\log x} dx$

$\frac{1}{\log x} > \frac{1}{x} \;\forall\; x > 1 \Rightarrow \frac{\sqrt{x}}{\log x} > \frac{1}{\sqrt{x}} \;\forall\; x > 1$

$\int_1^2 \frac{1}{\sqrt{x}} dx = 2\sqrt{x}\Big|_1^2 = 2\sqrt{2} - 2$ = finite $\Rightarrow$ convergent

$\int_1^2 \frac{\sqrt{x}}{\log x} dx$ may / may not be convergent

2nd Method: $f(x) = \frac{\sqrt{x}}{\log x}$ Let $g(x) = \frac{1}{x \log x}$

$$\lim_{x \to 1^+} \frac{f(x)}{g(x)} = 1\sqrt{1} = 1$$

Let $\log x = t$ $\quad x = 1 \Rightarrow t = 0$, $x = 2 \Rightarrow t = \log 2$

$\frac{1}{x} dx = dt$

$$\int_1^2 \frac{1}{x \log x} dx = \int_0^{\log 2} \frac{1}{t} \cdot dt$$

$$= \log t \Big|_0^{\log 2} = \text{infinite (diverge)}$$

$$\therefore \int_1^2 \frac{\sqrt{x}}{\log x} dx \text{ is also divergent.}$$

(GATE-00): $\lim_{a \to \infty} \int_1^a x^{-4} dx$

(a) diverges ✓(b) converges to $1/3$ c) converges to $-1/a^3$ d) converges to 0

Sol: $\lim_{a \to \infty} \int_1^a x^{-4} dx = \lim_{a \to \infty} \left[\frac{x^{-3}}{-3}\right]_1^a = \lim_{a \to \infty} \left[\frac{a^{-3}}{-3} - \frac{1}{-3}\right]$

$$= \lim_{a \to \infty} -\frac{1}{3a^3} + \frac{1}{3} = 0 + \frac{1}{3} = 1/3$$

(GATE-05): $\int_1^\infty x^{-3} dx$

$$= \frac{x^{-2}}{-2}\Big|_1^\infty = -\frac{1}{2}\left[\frac{1}{\infty} - 1\right] = 1/2$$

(GATE-08 ME): Which of the following integrals is unbounded?

(a) $\int_0^{\pi/4} \tan x \, dx$ (b) $\int_0^\infty \frac{1}{1+x^2} dx$ (c) $\int_0^\infty x e^{-x} dx$ ✓(d) $\int_0^1 \frac{1}{1-x} dx$

Sol: (a) $\int_0^{\pi/4} \tan x \, dx = \log \sec x \Big]_0^{\pi/4} = \log\sqrt{2} - \log 1 = \log\sqrt{2}$

(b) $\int_0^\infty \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_0^\infty = \tan^{-1}\infty - \tan^{-1} 0 = \pi/2 - 0 = \pi/2$

(c) $\int_0^\infty x e^{-x} dx = \left[x \cdot \frac{e^{-x}}{-1} - \frac{e^{-x}}{(-1)^2}\right]_0^\infty = 0 - 0 + \frac{1}{(-1)^2} = 1$

(d) $\int_0^1 \frac{1}{1-x} dx = \left[\frac{\log(1-x)}{-1}\right]_0^1 = \log 0 + \log 1 = \infty + 0 = \infty \Rightarrow$ unbounded

(GATE-10 ME): $\int_{-\infty}^\infty \frac{1}{1+x^2} dx \Rightarrow$ even function

$$= 2\int_0^\infty \frac{1}{1+x^2} dx = 2 \cdot \pi/2 = \pi$$

Gamma function: $\Gamma(n) = \int_0^{\infty} e^{-x} \cdot x^{n-1}\, dx \quad (n>0)$.

Note: (i) $\Gamma 1 = 1$

(ii) $\Gamma \tfrac{1}{2} = \sqrt{\pi}$

(iii) $\Gamma(n+1) = n\Gamma n \quad \forall\, n>0$

(iv) $\Gamma(n+1) = n! \quad \forall\, n \in Z^+$

(v) $\int_0^{\infty} e^{-ax} \cdot x^{n-1}\, dx = \frac{\Gamma n}{a^n}$

prob: $\int_0^{\infty} e^{-x^2} dx =$

Sol: Let $x^2 = t \Rightarrow 2x\, dx = dt \Rightarrow dx = \frac{1}{2} t^{-1/2}\, dt$

$$I = \int_0^{\infty} e^{-t} \cdot \frac{1}{2} t^{-1/2} dt = \frac{1}{2}\int_0^{\infty} e^{-t} \cdot t^{1/2-1} dt = \frac{1}{2}\Gamma\tfrac{1}{2} = \frac{\sqrt{\pi}}{2}$$

prob: $\int_{-\infty}^{\infty} e^{-x^2} dx = 2\int_0^{\infty} e^{-x^2} dx = 2 \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\pi}$

prob: $\int_0^1 (x \log x)^4\, dx =$

Sol: Let $\log x = -t \Rightarrow x = e^{-t} \Rightarrow dx = -e^{-t} dt$

$x=0 \Rightarrow t=-\infty$
$x=1 \Rightarrow t=0$

$$I = \int_{-\infty}^{0} [e^{-t} \cdot (-t)]^4 \cdot (-e^{-t} dt) = \int_0^{\infty} e^{-5t} \cdot t^{5-1} dt$$

$$= \frac{\Gamma 5}{5^5} = \frac{4!}{5^5}$$

$\log_a 0 = \begin{cases} \infty \,;\ a<1 \\ -\infty \,;\ a>1 \end{cases}$

Beta function: $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx \quad (m>0,\ n>0)$

Note: (i) $\beta(m,n) = \beta(n,m)$

(ii) $\beta(m,n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma m+n}$

(iii) $\beta(m,n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{n+m}} dx$

(iv) $\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cdot \cos^{2n-1}\theta\, d\theta$

i.e $\int_0^{\pi/2} \sin^p\theta \cdot \cos^q\theta \cdot d\theta = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

$(p>-\tfrac{1}{2},\ q>-\tfrac{1}{2})$

prob: $\int_0^2 x^7 (16-x^4)^{10} dx =$

Soln: Let $x^4 = 16t \Rightarrow 4x^3 dx = 16\, dt \Rightarrow x^3 dx = 4\, dt$

$x=0 \Rightarrow t=0$
$x=2 \Rightarrow 16 = 16t$
$t=1$

$$I = \int_0^1 16t\,(16-16t)^{10}\, 4\, dt = 16^{11} \cdot 4\int_0^1 t(1-t)^{10} dt$$

$$= 4 \times 16^{11} \times \beta(2,11)$$

$$= 4 \times 16^{11} \times \frac{\Gamma 2 \cdot \Gamma 11}{\Gamma 13} = 4 \times 16^{11} \times \frac{1 \times 10!}{12!}$$

prob: $\int_0^{\infty}\left(\frac{x}{1+x^2}\right)^3 dx$

sol: Let $x=\tan\theta \Rightarrow dx=\sec^2\theta\, d\theta$

$$I=\int_0^{\pi/2}\left(\frac{\tan\theta}{\sec^2\theta}\right)^3\sec^2\theta\, d\theta=\int_0^{\pi/2}\frac{\tan^3\theta}{\sec^4\theta}\cdot d\theta$$

$$=\int_0^{\pi/2}\sin^3\theta\cdot\cos\theta\, d\theta=\frac{1}{2}\beta\left(\frac{3+1}{2},\frac{1+1}{2}\right)=\frac{1}{2}\beta(2,1)$$

$$=\frac{1}{2}\times\frac{\Gamma 2\cdot\Gamma 1}{\Gamma 3}=\frac{1}{2}\times\frac{1\times 1}{2}=\frac{1}{4}$$

(GATE-94 ME): $\int_0^{\infty}e^{-y^3}\cdot y^{1/2}dy$

sol: put $y^3=t \Rightarrow 3y^2dy=dt$

$\Rightarrow t^{1/3}=y$ $\qquad y^{1/2}dy=\frac{1}{3}y^{-3/2}dt=\frac{1}{3}t^{-1/2}dt$

$$\int_0^{\infty}e^{-t}\cdot\frac{1}{3}t^{-1/2}dt=\frac{1}{3}\int_0^{\infty}e^{-t}\cdot t^{\frac{1}{2}-1}dt=\frac{1}{3}\Gamma(1/2)=\frac{\sqrt{\pi}}{3}$$

(GATE-10 PI): $\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-x^2/2}dx$ even function

sol: put $x^2/2=t \Rightarrow x\,dx=dt$

$\Rightarrow \sqrt{2t}\cdot dx=dt \Rightarrow dx=\frac{1}{\sqrt{2}}t^{-1/2}dt$

$$I=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-t}\frac{1}{\sqrt{2}}t^{-1/2}dt=\frac{1}{\sqrt{\pi}}\int_0^{\infty}e^{-t}\cdot t^{\frac{1}{2}-1}dt$$

$$=\frac{1}{\sqrt{\pi}}\cdot\Gamma(1/2)=\frac{\sqrt{\pi}}{\sqrt{\pi}}=1$$

MULTIPLE INTEGRALS:

double integral: An integral of the form $\boxed{\int_c^d \boxed{\int_a^b f(x)dx}\, dy}$ is called double integral. To evaluate this integral we have to follow the following steps.

1. First we evaluate the inner box & then the outer box.
2. while integrating w.r.to one variable we treat the other variable as constant
3. If the limits are constant limits then we need to follow the order in which dx & dy are present.
4. If the limits are variable limits then we need to follow the order in which the limits are present. and they are limits for that variable which is not present in the limits.

prob: $\int_{x=0}^{1}\int_{y=0}^{x^2}(x+y)\,dx\,dy = \int_{x=0}^{1}\left[\int_{y=0}^{x^2} x\cdot dy + \int_{y=0}^{x^2} y\,dy\right]dx$

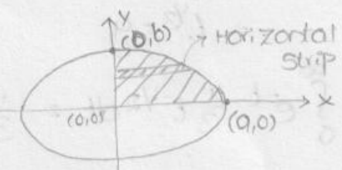
$$= \int_{x=0}^{1}\left[x\cdot(y)_0^{x^2} + \left(\frac{y^2}{2}\right)_0^{x^2}\right]dx$$

$$= \int_{x=0}^{1}\left[x^3 + \frac{x^4}{2}\right]dx = \left[\frac{x^4}{4} + \frac{x^5}{10}\right]_0^1 = \frac{1}{4} + \frac{1}{10}$$

prob: $\int_0^4\int_0^{x^2} e^{y/x}\cdot dy\cdot dx = \int_0^4\left[\frac{e^{y/x}}{1/x}\right]_0^{x^2} dx$

$$= \int_0^4 [xe^x - x]\,dx = (xe^x - e^x)_0^4 - \left(\frac{x^2}{2}\right)_0^4 = 3e^4 - 7$$

prob: The value of $\iint_R xy\,dx\,dy$. where R is a region in the +ve quadrant at the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is



sol: Consider horizontal strip

$x = 0$ to $x = \frac{a}{b}\sqrt{b^2-y^2}$

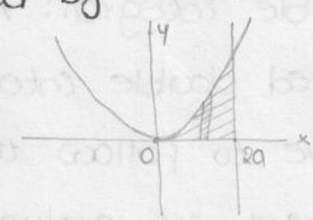
$y = 0$ to $y = b$

$$\iint_R xy\,dx\,dy = \int_{y=0}^{b}\int_{x=0}^{\frac{a}{b}\sqrt{b^2-y^2}} xy\cdot dx\cdot dy$$

$$= \int_{y=0}^{b} y\left[\frac{x^2}{2}\right]_0^{\frac{a}{b}\sqrt{b^2-y^2}}\cdot dy = \int_{y=0}^{b}\frac{y}{2}\left[\frac{a^2}{b^2}(b^2-y^2) - a^2\right]dy$$

$$= +\frac{a^2}{2}\int_{y=0}^{b}(y-y^3)\,dy = +\frac{a^2}{2}\left[\frac{y^2}{2} - \frac{y^4}{4}\right]_0^b = +\frac{a^2}{2}\left[\frac{b^4}{4}\right] = \frac{a^2b^2}{8}$$

prob: $\iint_R xy\,dx\,dy$ where R is the region bounded by x-axis $x = 2a$, $x^2 = 4ay$ $(a>0)$



sol: Consider vertical strip

$y = 0$ to $y = x^2/4a$, $x = 0$ to $2a$

$$\iint_R xy\,dx\,dy = \int_{x=0}^{2a}\int_{y=0}^{x^2/4a} xy\,dx\cdot dy$$

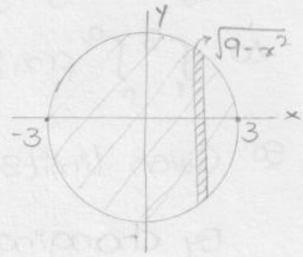
$$= \int_{x=0}^{2a} x\left[\frac{y^2}{2}\right]_0^{x^2/4a} dx = \int_{x=0}^{2a} x\cdot\frac{x^4}{32a^2}\,dx$$

$$= \frac{1}{32a^2}\int_{x=0}^{2a} x^5\,dx = \frac{1}{32a^2}\left(\frac{x^6}{6}\right)_0^{2a} = \frac{1}{32a^2}\cdot\frac{64a^6}{6} = \frac{a^4}{3}$$

Note: $\iint_R dx\,dy$ always represents area bounded by R

$\iiint_V dx\,dy\,dz$ always represents volume bounded by V.

Prob: $\iint_R dx\,dy$ where R is region bounded by the circle $x^2+y^2=9$



Sol: Consider vertical strip

$$y=-\sqrt{9-x^2} \text{ to } \sqrt{9-x^2}, \quad x=-3 \text{ to } 3$$

$$\iint_R dx\,dy = \int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dy \cdot dx$$

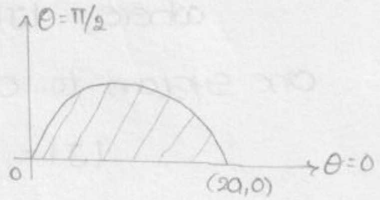
$$= \int_{-3}^{3}(y)_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx = \cancel{\int_{-3}^{3}\left(\frac{y^2}{2}\right)_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}}$$

$$= \int_{-3}^{3} 2\cdot\sqrt{9-x^2}\,dx = 2\cdot 2\int_0^3\sqrt{9-x^2}\,dx$$

$$= 4\left[\frac{x}{2}\sqrt{9-x^2}+\frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right)\right]_0^3 = 9\pi$$

Prob: The value of $\iint_R r^2\sin\theta\,dr\,d\theta$, where 'r' is the region bounded by the semi circle $r=2a\cos\theta$ above initial line is

Sol: $r=0$ to $2a\cos\theta$, $\theta=0$ to $\pi/2$



$$\iint_R r^2\sin\theta\,dr\,d\theta = \int_{\theta=0}^{\pi/2}\sin\theta\int_{r=0}^{2a\cos\theta} r^2\,dr\,d\theta$$

$$= \int_{\theta=0}^{\pi/2}\sin\theta\left(\frac{r^3}{3}\right)_0^{2a\cos\theta} d\theta$$

$$= \int_0^{\pi/2} 8a^3\frac{\cos^3\theta\cdot\sin\theta}{3}\,d\theta$$

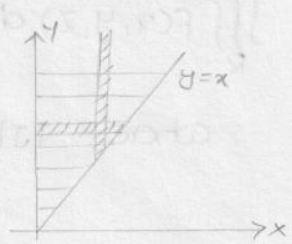
Let $\cos\theta = t \Rightarrow -\sin\theta\cdot d\theta = dt$ | $\theta=0 \Rightarrow t=1$, $\theta=\pi/2 \Rightarrow t=0$

$$= \int_1^0 \frac{8a^3}{3}t^3(-dt) = \frac{8a^3}{3}\int_0^1 t^3\,dt$$

$$= \frac{8a^3}{3}\left(\frac{t^4}{4}\right)_0^1 = \frac{8a^3}{3}\cdot\frac{1}{4} = 2a^3/3$$

Change of Order of integration:

Prob: $\int_0^\infty\int_x^\infty \frac{e^{-y}}{y}\,dy\,dx$



Sol: Given limits are $y=x$ to $y=\infty$
$x=0$ to $x=\infty$

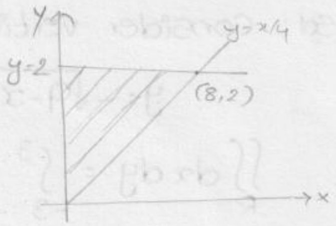
Horizontal strip $y=0$ to $y=\infty$, $x=0$ to y

$$\int_0^\infty\int_x^\infty \frac{e^{-y}}{y}\,dy\,dx = \int_{y=0}^{\infty}\left[\int_{x=0}^{y}\frac{e^{-y}}{y}\,dx\right]dy = \int_0^\infty e^{-y}\,dy$$

$$= -e^{-y} = [0+1] = 1$$

(GATE05)

prob: By changing the order of integration $\int_0^8 \int_{x/4}^{2} f(x,y)\,dy\,dx$ leads to $\int_r^s \int_p^q f(x,y)\,dx\,dy$ then q is



Sol: Given limits $y = x/4$ to 2, $x = 0$ to 8

By changing limits, $y = 0$ to 2, $x = 0$ to $4y$

$\therefore q = 4y$

prob: Evaluate $\int_0^2 \int_0^x \int_0^{\sqrt{x+y}} z\,dz\,dy\,dx$

Sol: $I = \int_0^2 \int_0^x \left[\frac{z^2}{2}\right]_0^{\sqrt{x+y}} dy\,dx = \int_0^2 \left[\frac{(x+y)^2}{4}\right]_0^x dx = 2$

Change of variables: In a double integral, if $x = f(u,v)$ & $y = g(u,v)$ then $\iint_R \phi(x,y) = \iint_R \phi(f,g)\,|J|\,du\,dv = \iint_R \phi(u,v)\,|J|\,du\,dv$

where, $|J| \rightarrow$ Jacobian of transformation used to transform one system to other.

$$|J| = J\left(\frac{x,y}{u,v}\right) = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

cartesian form $(x,y) \longrightarrow$ polar form (r,θ):

$x = r\cos\theta \qquad y = r\sin\theta$

$$|J| = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

$$\Rightarrow \iint_R \phi(x,y)\,dx\,dy = \iint_R \varphi(r,\theta)\,r\,dr\,d\theta$$

→ In a triple integral if $x = f(u,v,\omega)$, $y = g(u,v,\omega)$, $z = h(u,v,\omega)$

then $\iiint_R F(x,y,z)\,dx\,dy\,dz = \iiint_R \varphi(u,v,\omega)\,|J|\,du\,dv\,d\omega$

where $|J| = J\left(\frac{x,y,z}{u,v,\omega}\right) = \frac{\partial(x,y,z)}{\partial(u,v,\omega)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial \omega} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial \omega} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial \omega} \end{vmatrix}$

cartesian (x,y,z) to cylindrical polar form: (r,θ,z)

$x = r\cos\theta \qquad y = r\sin\theta \qquad z = z$

$$|J| = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

$$\iiint_R \phi(x,y,z)\,dx\,dy\,dz = \iiint_R \varphi(r,\theta,z)\, r\,dr\,d\theta\,dz$$

Cartesian to spherical polar form: (r,θ,ϕ)

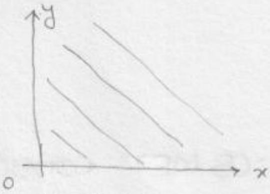
$x = r\sin\theta\cos\phi \qquad y = r\sin\theta\sin\phi \qquad z = r\cos\theta$

$$|J| = \begin{vmatrix} \sin\theta\cdot\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\cdot\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix} = r^2\sin\theta$$

$$\iiint_R \phi(x,y,z)\,dx\,dy\,dz = \iiint_R \varphi(r,\theta,\phi)\, r^2\sin\theta\,dr\,d\theta\,d\phi$$

(IN 07)

prob: $\int_0^\infty\int_0^\infty e^{-(x^2+y^2)}\,dx\,dy$



$r \to 0$ to ∞
$\theta \to 0$ to $\pi/2$

sol: Let $x = r\cos\theta$, $y = r\sin\theta$ then $|J| = r$

& $x^2+y^2 = r^2$

$$\int_0^\infty\int_0^{\pi/2} e^{-r^2}\cdot r\cdot dr\,d\theta$$

Let $r^2 = t \Rightarrow r\,dr = dt/2$

$$= \int_0^{\pi/2}\int_0^\infty e^{-t}\cdot dt/2\cdot d\theta = \int_0^{\pi/2}\frac{1}{2}\left[-e^{-t}\right]_0^\infty d\theta = \frac{1}{2}\int_0^{\pi/2} d\theta = \pi/4$$

(GATE 05)

Prob: By a change of variable $x(u,v) = uv$ & $y(u,v) = v/u$ in double integration, the integrand $f(x,y)$ changes to $f(uv, v/u)\,\phi(u,v)$ then $\phi(u,v)$ is

sol: $\phi(u,v) = |J| = \begin{vmatrix} v & u \\ -\frac{v}{u^2} & 1/u \end{vmatrix} = \frac{v}{u} + \frac{v}{u} = \frac{2v}{u}$

(GATE-95): By reversing the order of integration $\int_0^2\int_{x^2}^{2x} f(x,y)\,dy\,dx$ may be represented as

(a) $\int_0^2\int_{x^2}^{2x} f(x,y)\,dy\,dx$ (b) $\int_0^2\int_y^{\sqrt{y}} f(x,y)\,dx\,dy$ ✓(c) $\int_0^4\int_{y/2}^{\sqrt{y}} f(x,y)\,dx\,dy$ (d) $\int_{x^2}^{2x}\int_0^2 f(x,y)\,dy\,dx$

sol $y = x^2$ to $y = 2x$ $\qquad$ $x=0$ to $x=2$

$\downarrow \qquad\qquad \downarrow \qquad\qquad \downarrow \qquad\qquad \downarrow$

$x = \sqrt{y} \qquad x = y/2 \qquad y = 0^2 = 0 \qquad y = 2^2 = 4$

$\qquad\qquad\qquad\qquad = 2(0) = 0 \qquad = 2(2) = 4$

$$\therefore \int_0^2\int_{x^2}^{2x} f(x,y)\,dy\,dx = \int_{y=0}^{4}\int_{x=y/2}^{\sqrt{y}} f(x,y)\,dx\,dy$$

(GATE-04): The volume of an object expressed in spherical co-ordinate is given by $v = \int_0^{2\pi}\int_0^{\pi/3}\int_0^{1} r^2 \sin\phi \, dr\, d\phi\, d\theta$. The value of the integral

Sol: $v = \int_0^{2\pi}\int_0^{\pi/3}\left[\frac{r^3}{3}\right]_0^1 \sin\phi \, d\phi\, d\theta = \frac{1}{3}\int_0^{2\pi}(-\cos\phi)_0^{\pi/3} d\theta$

$= \frac{1}{3}\times\frac{1}{2}\times 2\pi = \pi/3$

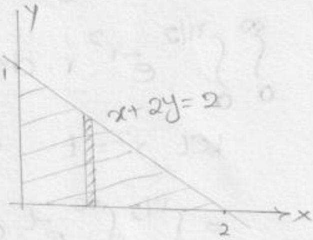
(GATE-08 CS): $\int_0^3\int_0^x (6-x-y)\,dx\,dy$

Sol: $\int_0^3 \left[6y - xy - y^2/2\right]_0^x dx = \int_0^3 (6x - x^2 - x^2/2)\,dx$

$= \int_0^3 (6x - 3x^2/2)\,dx = 6\left(\frac{x^2}{2}\right)_0^3 - \frac{3}{2}\left(\frac{x^3}{3}\right)_0^3$

$= 6\times\frac{9}{2} - \frac{3}{2}\times\frac{27}{3} = 27 - 13.5 = 13.5$

(GATE-08 ME): consider shaded triangular region P shown in the figure. what is $\iint_P xy\,dx\,dy$?



Sol: consider vertical strip

$x + 2y = 2$
$\Downarrow$
$y = \frac{2-x}{2}$

$y = 0$ to $\frac{2-x}{2}$

$x = 0$ to 2

$I = \int_0^2\int_0^{\frac{2-x}{2}} xy.\,dy\,dx = \int_0^2 x\left(\frac{y^2}{2}\right)_0^{\frac{2-x}{2}} dx$

$= \int_0^2 \frac{x}{2}\left(\frac{2-x}{2}\right)^2 dx = 1/6$

(GATE-14 EC): The volume under the surface $z(x,y) = x+y$ and above the triangle in the xy plane defined by $\{0\le y\le x$ and $0\le x\le 12\}$ is

Sol: volume $= \int_0^{12}\int_0^{x} z\,dy\,dx = \int_0^{12}\int_0^x (x+y)\,dy\,dx$

$= \int_0^{12}\left(xy + y^2/2\right)_0^x dx = \int_0^{12}\left(x^2 + x^2/2\right)dx$

$= \frac{3}{2}\left(x^3/3\right)_0^{12} = 864$

(GATE-14EE): To evaluate the double integral $\int_0^8\int_{y/2}^{(y/2+1)}\left(\frac{2x-y}{2}\right)dx\,dy$ we make the substitution $u = \left(\frac{2x-y}{2}\right)$ and $v = y/2$. The integral will reduce to

Sol: $v = y/2 \Rightarrow dv = \frac{1}{2}dy \Rightarrow dy = 2dv$

$u = \left(\frac{2x-y}{2}\right) \Rightarrow du = dx$.

AS $x: \frac{y}{2} \to \frac{y}{2}+1 \Rightarrow u: 0 \to 1$ and $y: 0 \to 8 \Rightarrow v: 0 \to 4$

$\therefore$ integral ~~becomes~~ become $\int_0^4 \int_0^1 2u\,du\,dv$

(GATE-14 ME): $\int_0^2 \int_0^x e^{x+y}\,dy\,dx$

sol: $\int_0^2 \int_0^x e^{x+y}\,dy\,dx = \int_0^2 e^x (e^y)_0^x\,dx = \int_0^2 e^x(e^x - 1)\,dx$

$= \int_0^2 (e^{2x} - e^x)\,dx = \left[\frac{e^{2x}}{2} - e^x\right]_0^2 = \frac{(e^2-1)^2}{2}$

Lengths, Areas & volumes:

1. Length of an arc of a curve $y = f(x)$ between the lines $x = x_1$ & $x = x_2$ is $l = \int_{x_1}^{x_2} \sqrt{1 + (dy/dx)^2}\,dx$

2. Length of an arc of a curve $x = f(t)$ & $y = g(t)$ between $t = t_1$ to $t = t_2$ is $l = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + (dy/dt)^2}$

3. Area of the region bounded by the curve $y = f(x)$ & $y = g(x)$ between $x = x_1$ & $x = x_2$ is $A = \int_{x_1}^{x_2} [g(x) - f(x)]\,dx = \int_{x_1}^{x_2} \int_{f(x)}^{g(x)} dy\,dx$

4. The volume generated by revolving $y = f(x)$ between $x = x_1$ & $x = x_2$ about x-axis is

$$V = \int_{x_1}^{x_2} \pi y^2\,dx$$

About y-axis is $V = \int_{y_1}^{y_2} \pi x^2\,dy$

5. Inpolar form: (i) about initial line (i.e $\theta = 0$) $V = \int_{\theta_1}^{\theta_2} \frac{2\pi}{3} r^3 \sin\theta\,d\theta$

(ii) about line $\theta = \pi/2$ $V = \int_{\theta_1}^{\theta_2} \frac{2\pi}{3} r^3 \cos\theta\,d\theta$

(GATE 08 ME)

prob: The length of the curve $y = \frac{2}{3}x^{3/2}$ b/w $x = 0$ & 1 is

sol: $\frac{dy}{dx} = \frac{2}{3} \cdot \frac{3}{2} x^{\frac{3}{2}-1} = x^{1/2}$

$l = \int_{x_1}^{x_2} \sqrt{1 + (dy/dx)^2}\,dx = \int_0^1 \sqrt{1+x}\,dx = \left[\frac{(1+x)^{3/2}}{3/2}\right]_0^1 = 1.22$

prob: The length of curve $x = \cos^3\theta$ $y = \sin^3\theta$ b/w $\theta = 0$ & $\pi/2$ is

sol: $l = \int_0^{\pi/2} \sqrt{(3\cos^2\theta(-\sin\theta))^2 + (3\sin^2\theta \cdot \cos\theta)^2}\,d\theta = \int_0^{\pi/2} 3\sin\theta\cos\theta\,d\theta$

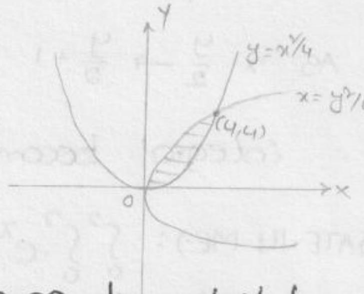
$= 3/2$.

(GATE 09 ME)

Prob: The area between the curve $y^2 = 4x$ & $x^2 = 4y$ is

Sol: $A = \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4}\right) dx = \left[2 \cdot \frac{x^{3/2}}{3/2} - \frac{1}{4} \cdot \frac{x^3}{3}\right]_0^4$

$= \left[\frac{4}{3} \times 8 - \frac{1}{4} \times \frac{64}{3}\right] = 16/3$

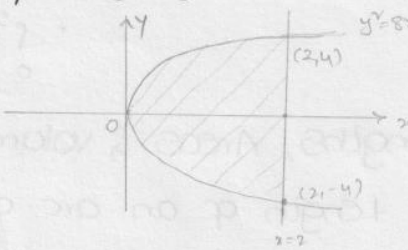


(GATE-94)

Prob: The volume generated by revolving the area bounded by parabola $y^2 = 8x$ and st. line $x = 2$ about y-axis is

Sol: $V = \int_{y_1}^{y_2} \pi x^2 dy = \int_{-4}^{4} \pi \cdot \frac{y^4}{64} dy$

$= \frac{\pi}{64}\left[\frac{y^5}{5}\right]_{-4}^{4} = \frac{32\pi}{5}$

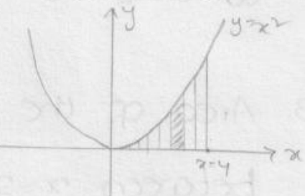


Total volume $= \int_{-4}^{4} \pi x^2 dy = \int_{-4}^{4} \pi (2)^2 dy = 32\pi$

Remaining volume $= 32\pi - 32\pi/5 = 128\pi/5$

(GATE-97): Area bounded by the curve $y = x^2$ and the lines $x = 4$ and $y = 0$ is given by

Sol: Consider vertical strip $y = 0$ to $y = x^2$
$x = 0$ to $x = 4$



$\int_{x=0}^{4} \int_{y=0}^{x^2} dx\,dy = \int_{x=0}^{4} (y)_0^{x^2} dx$

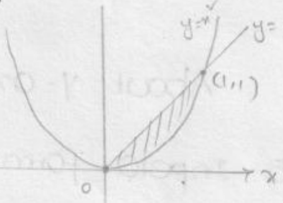
$= \int_0^4 x^2 dx = \left.\frac{x^3}{3}\right|_0^4 = \frac{4^3}{3} = 64/3$

(GATE-12 ME, PI)

(GATE-04): The area enclosed b/w the parabola $y = x^2$ and the straight line $y = x$ is

Sol: Area $= \int_0^1 (x - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1$

$= \frac{1}{2} - \frac{1}{3} = \frac{3-2}{6} = 1/6$



(GATE-10 ME): The parabolic arc $y = \sqrt{x}$ $1 \le x \le 2$ is revolved around the x-axis. The volume of the solid of revolution is

Sol: volume $= \int_{x_1}^{x_2} \pi y^2 dx = \int_1^2 \pi (\sqrt{x})^2 dx = \pi \left(\frac{x^2}{2}\right)_1^2 = 3\pi/2$